Factorial Linear Forms and Denominators Detailed Proofs and Rationality Criteria

Erdős Problem #68

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Abstract

We classify multipliers in finitely supported factorial-weighted cancellation forms $MS + k$, where $S = \sum_{n\geq 2}(n! - 1)^{-1}$, with integer coefficients on $n \geq 2$. An integral basis reduces cancellation through $D \geq 2$ and exclusion of index one to a Diophantine equation whose coefficient gcd is determined below $2D^2$. Examples separate least multiplier, support size and coefficient norm. We prove a lower bound for common-denominator growth, analyse prime-power cancellation on reduction, and derive equivalent carry and factorial-digit rationality criteria. Recorded computations exclude $q \mid 299999!$ and $q < 2^{39990}$ under $S = a/q$. These computations are outside Lean; the required nonintegrality at arbitrarily large indices remains unproved.

Problem 0.1 (Erdős #68)**.** Is $S = \sum_{n\geq 2}(n! - 1)^{-1}$ irrational?

Erdős posed this question in 1988, together with the expectation that the corresponding series with denominators $n! + t$ should be transcendental for every integer t [1, p. 102]. For negative t, the sum begins after all zero or negative denominators. Any further change of starting index adds a rational number. The expectation for all integer shifts remains unproved here. The problem number follows Bloom's catalogue [2].

For finitely supported integer vectors on $n \geq 2$, cancellation forms $MS + k$ give an explicitly computable ideal of multipliers. Correcting adjacent factorial differences at proper divisors gives an integral basis. Cancellation fixes its lower coordinates; excluding index one leaves a single Diophantine equation. Its gcd, computed below $2D^2$, gives the least positive multiplier and an attaining vector. At depth four, the support restriction strengthens $115 \mid M$ to $1380 \mid M$. This classification is independent of nonintegrality.

The accompanying short paper proves this classification and bounds one resulting remainder between consecutive integers. Here we develop the support and coefficient-norm refinements and a primitive construction on arithmetic progressions. These solve different problems: at fixed M, changing the vector changes $MS+k$ only by an integer, so it cannot alter nonintegrality. The progression construction makes M divisible by every fixed denominator eventually, but still needs a gap estimate for those same forms.

Authorship and AI use. Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

The later sections study that arithmetic difficulty from two directions. Common-denominator growth measures the cost of clearing terms; prime-power calculations determine what survives reduction. Carries and factorial digits instead characterize rationality itself. The resulting equivalences identify what an irrationality proof must show, without proving that the required departures occur arbitrarily far out.

Throughout, write

$$d_n = n! - 1, \qquad H_N = \sum_{n=2}^{N} \frac{1}{d_n}, \qquad L_N = \operatorname{lcm}(d_2, \dots, d_N), \qquad A_N = \sum_{n=2}^{N} \frac{L_N}{d_n}.$$

Thus $H_N = A_N/L_N$. We write $\operatorname{den}(x)$ for the positive reduced denominator of a rational number x. Under $S = a/q$, every form $MS + k$ with $M, k \in \mathbb{Z}$ and $q \mid M$ is integral. For each positive integer q, an irrationality argument of this kind must produce a nonintegral form with $q \mid M$. Both properties must hold for the same vector; cancellation and growth of M alone do not establish this.

The proof of the classification is in §1.1; the following table locates the additional arguments. In particular, the lower bound for L_N concerns the unreduced common denominator, not automatically $\operatorname{den}(H_N)$. The appendices retain the supplementary identities, limits of the attempted arguments, and formal-source concordance.

Question	Where it is treated
Which M and coefficient vectors are possible?	§1.1
What do support and norm minimisation change?	§1.2
Which prime powers survive reduction?	§2
How fast does the common denominator grow?	§3
How does growth constrain the support parameter?	§3.1
Which eventual patterns characterize rationality?	§4
Which denominators are excluded by finite computations?	§6
Where are the formal declarations?	Appendix A.1

1 Integer coefficient vectors and cancellation

We need weights that preserve the fractional part of $i!/(d!-1)$ and whose scaled adjacent differences have a simple form. Both requirements are met by removing one factor $d!$ for each complete block of d indices. For a finitely supported integer vector $c = (c_i)$ on $i \geq 2$, put

$$M(c) = \sum_i c_i\, i!, \qquad W_{d,i} = \frac{i!}{(d!)^{\lfloor i/d \rfloor}}, \qquad V_d(c) = \sum_i c_i W_{d,i} \quad (d \geq 2).$$

We allow either sign for $M(c)$ and for $V_d(c)$. The associated remainder is

$$\mathcal{R}(c) = \sum_{d \geq 2} \frac{V_d(c)}{d! - 1}.$$

The index i selects a coefficient, whereas d selects a summand of the remainder. If c is supported on $i \le N$, then for $d > N$ all floor exponents vanish and $V_d(c) = M(c)$. The series therefore converges absolutely. Cancellation of the first weighted sums leaves a finite signed sum followed by $M(c)$ times the original tail. We show below that the remainder differs from $M(c)S$ by an integer.

The chosen exponent is a floor, rather than the largest possible power of $d!$ dividing $i!$. For example, at $d = 2$, $i = 6$, it is 3 although $2^4 \mid 6!$. What matters is that it changes between $i - 1$ and i exactly when $d \mid i$. This makes adjacent differences amenable to divisor elimination in §1.1. Each $W_{d,i}$ is an integer: writing $i = kd + r$ with $0 \le r < d$, the quotient $i!/((d!)^k r!)$ is a multinomial coefficient. Since $i! = (d!)^{\lfloor i/d \rfloor} W_{d,i}$ and $d! \equiv 1 \pmod{d! - 1}$,

$$V_d(c) \equiv M(c) \pmod{d! - 1}. \tag{1}$$

Theorem 1.1 (divisibility of the difference)**.** *For every finite integer support and every $d \ge 2$ there is an integer k with $V_d(c) = M(c) + (d! - 1)k$.*

Proof. By (1), $d! - 1$ divides $V_d(c) - M(c)$. Thus $k = (V_d(c) - M(c))/(d! - 1)$ is an integer and gives the identity. The modulus is positive also at $d = 2$, when it equals one. □

For any $N \ge 2$ at least as large as every supported index, the terms with $d > N$ equal $M(c)/(d! - 1)$. Therefore

$$\mathcal{R}(c) - M(c)S = \sum_{d=2}^{N} \frac{V_d(c) - M(c)}{d! - 1} \in \mathbb{Z}.$$

The sum is finite and each term is integral by (1). This argument also holds when index 1 is temporarily allowed in the basis calculation below. In particular, $M = 0$ gives an integral remainder, and fixing $M(c)$ fixes its fractional part. These conclusions hold for every finite vector, before imposing cancellation.

By (1), a vanishing d-th weighted sum forces $(d! - 1) \mid M(c)$, and annihilating every weighted sum $2 \le d \le D$ forces $L_D \mid M(c)$, where $L_D = \operatorname{lcm}_{2 \le d \le D}(d! - 1)$ is the same quantity as in §3. With M fixed, the same congruence fixes the fractional part of each quotient $V_d(c)/(d! - 1)$ separately.

1.1 An integral basis and the possible values of M

Fix $D \ge 2$. We temporarily allow a coefficient at index one so that the adjacent-difference vectors form a triangular basis. After solving $V_2 = \cdots = V_D = 0$, we impose one further equation to remove that coordinate. Only the coefficient space is enlarged; the series S continues to start at $d = 2$, with no term $1/(1! - 1)$.

Let e_n be the unit vector at index n. For $n \ge 2$ put

$$T_n = n e_{n-1} - e_n.$$

We have $M(T_n) = 0$. If $d \nmid n$, the two floor exponents in its weighted sum agree, and $nW_{d,n-1} = W_{d,n}$. If $d \mid n$, the exponent increases by one, so $nW_{d,n-1} = d! W_{d,n}$. Hence

$$V_d(T_n) = (d! - 1) W_{d,n} \mathbf{1}_{d \mid n}.$$

For example, $T_4 = 4e_3 - e_4$ has weighted sums $V_2 = 6$ and $V_4 = 23$, and all its other weighted sums vanish. Since $T_2 = 2e_1 - e_2$ has only $V_2 = 1$ nonzero, $T_4 - 6T_2$ isolates the fourth weighted sum.

We therefore define, in increasing order of n,

$$U_n = T_n - \sum_{\substack{d \mid n \\ 2 \le d < n}} W_{d,n} U_d.$$

Assume the construction has isolated the weighted sums below n. Subtracting $W_{d,n}U_d$ removes the contribution at a proper divisor d and changes no other weighted sum. At $d = n$ we have $W_{n,n} = 1$. Induction therefore gives $M(U_n) = 0$ and $V_d(U_n) = (d! - 1)\mathbf{1}_{d=n}$ for every $d \ge 2$.

For the basis claim, restrict to indices $1, \dots, N$. The matrix with columns $e_1, T_2, \dots, T_N$ is triangular over $\mathbb{Z}$ with diagonal $1, -1, \dots, -1$. The change from T_n to U_n is triangular with diagonal one. Both determinants are units in $\mathbb{Z}$, so both inverses have integer entries. In back-substitution the only divisors are 1 and -1; no rational coefficients are needed. Thus $e_1, U_2, \dots, U_N$ represents every integer vector uniquely. If this expansion is $be_1 + \sum z_d U_d$, then $M(c) = b$ and $V_d(c) = b + (d! - 1)z_d$. These identities determine the coordinates:

$$c = M(c)e_1 + \sum_{d \ge 2} \frac{V_d(c) - M(c)}{d! - 1} U_d.$$

The congruence (1) also verifies the integrality of the displayed coefficients. They vanish for $d > N$, since then $V_d(c) = M(c)$, so the expansion is finite. These same coefficients occur in the integer-difference identity at the start of §1: their sum is $\mathcal{R}(c) - M(c)S$. Thus the coordinates also recover the integer term in the linear form.

Cancellation through D forces $L_D \mid M(c)$. To obtain $M(c) = L_D$ in the enlarged space, start with $L_D e_1$, whose weighted sums all equal L_D. The vector U_d changes only the dth sum, by $d! - 1$. Hence define

$$K_D = L_D e_1 - \sum_{d=2}^{D} \frac{L_D}{d! - 1} U_d.$$

This vector satisfies $M = L_D$ and $V_2 = \cdots = V_D = 0$. All solutions in the enlarged coefficient space consequently have the unique form

$$c = tK_D + \sum_{n > D} z_n U_n, \qquad M(c) = tL_D, \tag{2}$$

where t, z_n are integers and only finitely many z_n are nonzero. The only forbidden coefficient is now the one at index 1. Write $a_D = (K_D)_1$ and $u_n = (U_n)_1$. Removing this coefficient is equivalent to

$$ta_D + \sum_{n > D} z_n u_n = 0. \tag{3}$$

The free higher basis coefficients can change the first coefficient by every multiple of $\gcd\{u_n : n > D\}$, and only by such a multiple. Thus the gcd gives the exact solvability

condition. We next show how to compute it using a bounded list of coefficients. Here $u_2 = 2$, whereas for $n > 2$ the recurrence reads

$$u_n = -\sum_{\substack{d \mid n \\ 2 \le d < n}} W_{d,n} u_d.$$

It gives $u_n = 0$ for odd $n > 1$. If p is prime, the only proper divisor of $2p$ with a nonzero contribution is 2, and therefore $u_{2p} = -(2p)!/2^{p-1}$, including $p = 2$.

Choose a prime p with $D/2 < p \le D$. Such a prime exists by Bertrand's postulate when $D \ge 3$, and $p = 2$ works when $D = 2$. The coefficient u_{2p} lies in the tail and divides $(2p)!$. Once a finite initial interval includes that coefficient, its gcd therefore divides this factorial. To control the small-divisor weights beyond the interval, we shall require $n/d \ge 2p$ whenever $d \le D$. This leads to the cutoff $H = D(2p-1)$. We claim that

$$\gcd\{u_n : n > D\} = \gcd(u_{D+1}, \dots, u_H) > 0, \qquad H < 2D^2. \tag{4}$$

Let g be the finite gcd on the right. Its value could in principle decrease if we included later coefficients. To rule this out, we prove $g \mid u_n$ for every $n > D$ by strong induction. The index $2p$ lies in the finite interval and has a nonzero coefficient, so $g > 0$ and $g \mid (2p)!$. The definition proves the induction claim through H.

For $n > H$, each term of the recurrence is $W_{d,n}u_d$ with $d < n$. If $d > D$, divisibility comes from u_d by induction. If $d \le D$, we cannot use that hypothesis and instead use the weight. Put $k = n/d$. The cutoff was chosen so that $n > D(2p-1)$ forces $k \ge 2p$ for every such divisor. Also,

$$\frac{W_{d,n}}{k!} = \frac{n!}{(d!)^k k!} \in \mathbb{Z},$$

because the quotient counts partitions of n labelled objects into k unordered blocks of size d. Hence $g \mid (2p)! \mid k! \mid W_{d,n}$. Thus every product in the recurrence is divisible by g, completing the induction. The finite gcd divides the entire tail, and inclusion of the finite interval gives the reverse divisibility. Finally, $H \le D(2D-1) < 2D^2$.

Theorem 1.2 (the set of attainable values of M)**.** *Fix $D \ge 2$ and a prime p with $D/2 < p \le D$. Put*

$$H = D(2p-1), \qquad G_D = \gcd\{|u_n| : D < n \le H\}, \qquad \mu_D = L_D \frac{G_D}{\gcd(G_D, a_D)}.$$

For finite integer vectors supported on $n \ge 2$ and cancelling all weighted sums $2, \dots, D$, the possible values of M are exactly $\mu_D\mathbb{Z}$. The integer μ_D is positive, is independent of the eligible prime p, and is attained by a vector of coefficients with gcd one, that is, by a primitive vector.

Proof. Equation (4) identifies G_D with the gcd of all u_n for $n > D$. Their finite integer combinations are exactly $G_D\mathbb{Z}$. Hence (3) has a solution if and only if $G_D \mid ta_D$, or equivalently $G_D/\gcd(G_D, a_D) \mid t$. Substitution in (2) gives the claimed set of values of M. Bézout's identity applied to the finite list through H supplies a vector attaining its least positive element. A common coefficient divisor larger than one could be divided out, giving a smaller positive attainable value of M. The attaining vector is therefore primitive. Finally, the set of values of M is intrinsic to the cancellation equations, so its positive generator is independent of the eligible prime p. □

Fix $M \in \mu_D\mathbb{Z}$ and put $t = M/L_D$. The support equation (3) determines which basis coefficients are admissible. The following formula records their integer contribution and shows why changing the vector at that value of M cannot alter nonintegrality.

Theorem 1.3 (how the coefficient choices change the remainder)**.** *For the vector in* (2)*,*

$$\mathcal{R}\left(tK_D + \sum_{n>D} z_n U_n\right) = tL_D(S - H_D) + \sum_{n>D} z_n.$$

The remainder series converges for every finite vector supported away from index zero. A vector with $M = 0$ has integral remainder, and any two finite vectors with the same value of M have remainders differing by an integer.

Proof. For $d > D$, we have $V_d(K_D) = L_D$ and $V_d(U_n)/(d! - 1) = \mathbf{1}_{d=n}$. Summing these identities gives $tL_D(S - H_D) + \sum_{n>D} z_n$, as claimed. The first series converges and the second sum is finite.

Absolute convergence and the general integer-difference identity were proved at the start of §1, including for the auxiliary index 1. Taking $M = 0$ proves integrality; subtracting the identities for two vectors with equal M proves the last assertion. □

Here $M = tL_D$, so every denominator in H_D divides M. Thus $MH_D \in \mathbb{Z}$, and the integer in the original form $MS + k$ is $k = -MH_D + \sum_{n>D} z_n$.

1.2 Examples and support refinements

For $D = 4$ we need a common divisor of the entire coefficient tail, as well as a finite combination attaining it. Put $\Lambda_n = \operatorname{lcm}(1, \dots, n)$. We first prove $\Lambda_n \mid u_n$. For $d \mid n$ and a prime r, Legendre's formula gives

$$v_r(W_{d,n}) = \sum_{j\geq 1} \left(\left\lfloor \frac{n}{r^j} \right\rfloor - \frac{n}{d} \left\lfloor \frac{d}{r^j} \right\rfloor \right).$$

Every summand is nonnegative. If $d < r^j \leq n$, the second floor is zero and the first is at least one. Counting these powers gives $v_r(W_{d,n}) \geq v_r(\Lambda_n) - v_r(\Lambda_d)$, and therefore $\Lambda_n \mid W_{d,n}\Lambda_d$. Starting from $u_2 = 2$, induction in the recurrence proves $\Lambda_n \mid u_n$.

Thus $60 \mid u_n$ for all $n > 4$. Conversely, the finite calculation $u_6 = -180$, $u_8 = -4200$ gives $23u_6 - u_8 = 60$. The gcd is consequently $G_4 = 60$. Since $L_4 = 115$ and $a_4 = -55$, we obtain

$$\mu_4 = 115\frac{60}{\gcd(60, 55)} = 1380.$$

The vector $12K_4 + 253U_6 - 11U_8$ attains this value: its coefficient at index 1 is $12(-55) + 253(-180) - 11(-4200) = 0$.

The five-term vector used in the short paper is obtained from the same coordinates. The equality $12(-55) - 27(-180) - 4200 = 0$ first gives $12K_4 - 27U_6 + U_8$. Its coefficients at indices four and seven are removed by $-26U_5 + 8U_7$, respectively. Both corrections leave the first coordinate zero. Consequently

$$\begin{aligned} 12K_4 - 26U_5 - 27U_6 + 8U_7 + U_8 = {} & 1482e_2 - 784e_3 - 136e_5 \\ & + 83e_6 - e_8. \end{aligned} \tag{5}$$

The coefficients of the U_n in this expansion sum to -44. Since $H_4 = 143/115$, its remainder is $1380(S - H_4) - 44 = 1380S - 1760$. The exact interval calculation appears in §B.6. Thus the coordinates give both the original linear form and the cancellations, independently of a search over supports.

If support is additionally restricted to $n \le 6$, only $u_5 = 0$ and $u_6 = -180$ remain available in (3). The least positive value of M is then $115 \cdot 180/\gcd(180, 55) = 4140$. Thus a restriction on the largest supported index can increase the least attainable value of M. The following refinements specify the support bound in each case.

There is also a direct congruence proof using only the second and fourth weighted sums. For every $n \ge 2$,

$$1380 \mid 11n! - 46W_{2,n} + 12W_{4,n}. \tag{6}$$

The expression is zero for $n = 2, 3, 4$. For $n \ge 4$, it is divisible by 12: $12 \mid n!$ and $6 \mid W_{2,n}$, as follows from $W_{2,2r} = r! \prod_{j=1}^{r} (2j - 1)$ for $r \ge 2$ and the corresponding odd-index recurrence. For $n \ge 5$, it is divisible by 5 because $5 \mid n!$ and the denominators removed to form $W_{2,n}$ and $W_{4,n}$ are coprime to 5. Modulo 23, $n! \equiv W_{4,n}$ and $46W_{2,n} \equiv 0$. This proves the congruence. Consequently $V_2 = V_4 = 0$ forces $1380 \mid M$, without a restriction on the largest supported index. Conversely, $1482e_2 - 784e_3 - 136e_5 + 83e_6 - e_8$ satisfies $M = 1380$ and $V_2 = V_3 = V_4 = 0$. Thus imposing the additional equation $V_3 = 0$ does not change the possible values of M, although $3e_2 - e_3$ shows that it changes the set of admissible vectors.

Among integer vectors supported in $\{2, \dots, 8\}$ with $M = 1380$ and $V_2 = V_3 = V_4 = 0$, this vector has the smallest support. Every such vector has the following coefficients for some $t, x, y \in \mathbb{Z}$. The coefficient of K_4 remains $M/L_4 = 12$; here t is a new free parameter.

$$\begin{aligned} \lambda_2 &= 1482 - 4200t, & \lambda_3 &= -784 + 2240t, \\ \lambda_4 &= -680 + 1890t - 5x, & \lambda_5 &= x, \\ \lambda_6 &= 83 - 238t - 7y, & \lambda_7 &= y, \\ \lambda_8 &= -1 + 3t. \end{aligned} \tag{7}$$

Indeed, the values of the expression in (6) at indices $2, \dots, 8$ are

$$(0, 0, 0, 0, 4140, 28980, 328440).$$

Therefore the dual identity and $M = 1380$ give $11 = 3\lambda_6 + 21\lambda_7 + 238\lambda_8$. Modulo 3 this yields $\lambda_8 = -1 + 3t$, then $\lambda_6 + 7\lambda_7 = 83 - 238t$. Set $x = \lambda_5$ and $y = \lambda_7$; solving $V_2 = V_3 = V_4 = 0$ for the coefficients at indices $2, 3, 4$ gives (7). The coefficient matrix is $\left(\begin{smallmatrix} 1 & 3 & 6 \\ 2 & 1 & 4 \\ 2 & 6 & 1 \end{smallmatrix}\right)$, with determinant 55, and substitution verifies the converse. Indices $2, 3, 8$ are nonzero for every integral t; neither the pair (λ_4, λ_5) nor (λ_6, λ_7) can vanish together. Thus five nonzero entries are necessary at this support bound.

The smallest coefficient ℓ^1 norm for support in $\{2, \dots, 8\}$ is 2416, attained by $(\lambda_2, \dots, \lambda_8) = (1482, -784, 0, -136, -1, 12, -1)$. For $t \ge 1$, the absolute values at indices $2, 3, 8$ alone sum to $6443t - 2267 \ge 4176$; for $t \le -1$ they sum to $2267 - 6443t \ge 8710$. Thus a minimizer has $t = 0$. The remaining independent minima are $\min_x(|-680 - 5x| + |x|) = 136$ at $x = -136$ and $\min_y(|83 - 7y| + |y|) = 13$ at $y = 12$. For the first, $680 \le |-680 - 5x| + 5|x|$ is at most five times the objective. For the second, the objective

is at least 83 when $y \le 0$, at least 17 when $0 \le y \le 11$, and equals $8y - 83 \ge 13$ when $y \ge 12$. Adding the fixed contribution $1482 + 784 + 1 = 2267$ gives 2416. Equality forces $t = 0$, $x = -136$ and $y = 12$, so this norm minimizer is unique. Among the five-term vectors, $t = 0$ is again necessary for the least norm; the pair (λ_6, λ_7) can then have one nonzero entry only when $y = 0$, since $7 \nmid 83$. The pair (λ_4, λ_5) has one nonzero entry only at $x = 0$ or $x = -136$, and the latter is smaller. Hence the displayed five-term vector is the unique least-norm vector among those of minimum support, with norm 2486.

The additional equation $V_3 = 0$ changes the minimum support size and coefficient norm, while leaving the possible values of M unchanged. With only $V_2 = V_4 = 0$, group the seven coefficients by $a = \lambda_2 + 3\lambda_3$, $b = \lambda_4 + 5\lambda_5$, $c = \lambda_6 + 7\lambda_7$ and $d = \lambda_8$. Solving $V_2 = V_4 = 0$ and $M = 1380$ gives

$$(a, b, c, d) = (-870 + 2520t, -680 + 1890t, 83 - 238t, -1 + 3t), \qquad t \in \mathbb{Z}.$$

Each coordinate is nonzero for integral t, so four nonzero coefficients are necessary and sufficient. One four-term witness is $-290e_3 - 136e_5 + 83e_6 - e_8$, with norm 510. For the coefficient norm on the same set of indices, $|\lambda_2| + |\lambda_3| \ge |a|/3$ and $|\lambda_4| + |\lambda_5| \ge |b|/5$; when $t \ne 0$, these bounds together with $|d|$ already sum to at least 794. At $t = 0$, the four groups have unique norm minima 290, 136, 13 and 1, respectively. Thus the unique norm minimizer under only $V_2 = V_4 = 0$ is $-290e_3 - 136e_5 - e_6 + 12e_7 - e_8$, with norm 440. For comparison, with $M = 1380$ and support contained in $\{2, \dots, 8\}$, the two optimization problems give:

Cancellation	Minimum support size	Minimum ℓ^1 norm
$V_2 = V_4 = 0$	4	440
$V_2 = V_3 = V_4 = 0$	5	2416

In both rows the norm minimizer has one more nonzero coefficient than a support minimizer. Thus even with M and the allowed indices fixed, least support and least norm need not coincide. These are finite optimization statements, not irrationality conclusions.

There is a uniform restriction $12L_D \mid M$ on support $n \ge 2$. For $n = 2, 3$ one has $n! = 2W_{2,n}$. For $n \ge 4$, both $n!$ and $2W_{2,n}$ are divisible by 12: if $n = 2k$ or $2k + 1$, then $W_{2,2k} = k! \prod_{j=1}^{k} (2j - 1)$ is divisible by 6 for $k \ge 2$. Thus $12 \mid M(c) - 2V_2(c)$. Every $d! - 1$ is coprime to 6, so $V_2 = \cdots = V_D = 0$ implies $12L_D \mid M(c)$. This factor is attained at $D = 2$ and $D = 3$ by $-6e_2 + e_4$ and $-6e_2 - 8e_3 + 5e_4$, respectively. The next example shows that this need not be the whole restriction.

At $D = 6$, the same recurrence gives

$$L_6 = 9839515, \qquad a_6 = -2242555, \qquad G_6 = 840.$$

To verify the last value, $u_7 = 0$, while $840 = \operatorname{lcm}(1, \dots, 8)$ divides every u_n for $n \ge 8$ by the preceding divisibility. Conversely, $u_8 = -4200$, $u_{12} = -14386680$ and $\gcd(u_8, u_{12}) = 840$. Thus $\gcd(G_6, a_6) = 35$ and the least positive value of M is $\mu_6 = 24L_6$, not $12L_6$. It is attained by $24K_6 - 16240U_8 + U_{12}$: the coefficient at index 1 is zero and the coefficient at 12 is -1. The finite-gcd formula thus detects the extra factor at depth six, which the uniform divisibility by $12L_D$ misses.

1.3 Support bounds and nonintegrality

There are three separate choices: the vector, its multiplier M, and the cutoff used to estimate its remainder. The basis and support equations classify the first two. Once M is fixed, $\mathcal{R}(c) \equiv M(c)S \pmod{\mathbb{Z}}$ fixes the fractional part: integer translation may put a representative near zero but cannot change its distance from an integer.

To exclude every rational denominator, each $q > 0$ must divide the multiplier of some nonintegral form. Growth of the compulsory divisor $12L_D$ alone is insufficient: its 3-adic valuation is exactly one for every D, so it does not even force $9 \mid M$. By contrast, extending the cutoff changes neither M nor the remainder, only the finite enclosure used to test it.

For $c = -6e_2 + e_4$, we have $M(c) = 12$ and $\mathcal{R}(c) = 12S - 17$. At cutoff 4, its finite part is $-239/115$: the gap $9/115$ is smaller than the tail bound $24/119$. At cutoff 5, the finite part is $-27061/13685$, whose gap $13376/13685$ exceeds the new bound $24/719$. The finite part has crossed -2, so the next integer is now -1 rather than -2. This explains the larger gap. The second cutoff certifies nonintegrality of the same remainder; failure at the first cutoff was inconclusive. The successful test excludes denominators dividing the unchanged value $M = 12$, already covered by the finite exclusions.

For fixed c with $M(c) > 0$, the finite part increases to $\mathcal{R}(c)$ with omitted tail $M(c)(S - H_N)$. If $\mathcal{R}(c) \notin \mathbb{Z}$, its gap to the next integer is positive, whereas $2M(c)/((N+1)! - 1)$ tends to zero; the test therefore succeeds at every sufficiently large cutoff. If $\mathcal{R}(c) \in \mathbb{Z}$, the eventual gap equals the omitted tail and is smaller than this upper bound. Eventual success is consequently equivalent to nonintegrality of this fixed form, not an independent reason for it. The short paper's remainder comparison gives the inequality explicitly. For each $q > 0$ we still need a vector with $q \mid M(c)$ for which it succeeds; the successful cutoff need not be the largest supported index.

The support itself has a separate geometric restriction. Cancellation with $M \neq 0$ must cross a change in the relevant floor exponent; otherwise the weighted sum is proportional to M.

Theorem 1.4 (constant values of the floor in the weights)**.** *Let $d \geq 2$ and $k \geq 0$, and suppose every supported index i satisfies $kd \leq i < (k+1)d$. Then $M(c) = (d!)^k V_d(c)$. In particular, cancellation on the interval $d \leq i < 2d$ forces $M(c) = 0$; and if every supported index is at least d while $M(c) \neq 0$ and $V_d(c) = 0$, then some supported index is at least $2d$.*

Proof. On this interval the quotient $\lfloor i/d \rfloor$ is the constant k, so $i! = (d!)^k W_{d,i}$ for every supported index and the factor $(d!)^k$ comes out of the sum. The two consequences follow by taking $k = 1$ and by contraposition. □

The constant-floor condition is local. A quantitative bound also needs control of the multiplier: cancellation alone does not give $0 < M < (R+1)! - 1$. After proving the common-denominator estimates, §3.1 derives the resulting bounds on R.

At a prime index, the two-term vector T_p already changes only one weighted sum; there are no proper divisors to eliminate.

Theorem 1.5 (changing just one weighted sum)**.** *Let $p \geq 3$ be prime and let $c_{p-1} = p$, $c_p = -1$, with every other coefficient zero. Then $M(c) = 0$, $V_p(c) = p! - 1$, and $V_d(c) = 0$ for every $d \geq 2$ with $d \neq p$.*

Proof. Since $p \geq 3$, both indices $p-1$ and p lie in the prescribed coefficient domain $i \geq 2$. Here $M = p\,(p-1)!-p! = 0$. For $2 \leq d < p$ the quotient identity $\lfloor (p-1)/d \rfloor = \lfloor p/d \rfloor$ holds, since $d \nmid p$, so the two factorial weights carry the same power of $d!$ and the weighted sum evaluates to $p\,(p-1)!/(d!)^{\lfloor p/d \rfloor} - p!/(d!)^{\lfloor p/d \rfloor} = 0$. For $d > p$ both indices lie below d, so both weights are the plain factorials and the same cancellation occurs. At $d = p$ the weights are $(p-1)!$ and $p!/p! = 1$, giving $p\,(p-1)! - 1 = p! - 1$. □

For $p = 2$, the same auxiliary identities would require $c_1 = 2$. They do not define an admissible vector on the domain $i \geq 2$, since every vector supported on that domain has $c_1 = 0$.
Adding an integer multiple of this vector changes V_p by that multiple of $p! - 1$, without changing M or any other V_d. Suppose $M(c) \neq 0$, $V_2(c) = \cdots = V_D(c) = 0$, and every supported index exceeds a prescribed integer B. Choose a prime $p > \max(D, B+1, 2)$ and replace c by

$$c - \lfloor \mathcal{R}(c) + 1/2 \rfloor T_p.$$

Both new indices, $p-1$ and p, exceed B. Since $p > D$, the prescribed weighted sums remain zero, and M is unchanged. The new remainder is $\mathcal{R}(c) - \lfloor \mathcal{R}(c) + 1/2 \rfloor \in [-1/2, 1/2)$. The progression construction below supplies such an initial vector: choose its first index $r > B$.

An integral remainder rounds to zero; a nonintegral one rounds to a nonzero value. Thus rounding controls size but does not prove nonintegrality or exclude further denominators at fixed M. Moreover, $\lfloor \mathcal{R}(c) + 1/2 \rfloor$ depends on the unknown remainder: this is an existence formula for a representative, not an independent certificate of its nonvanishing.

1.4 A primitive solution on an arithmetic progression

We now separate two tasks that the basis classification alone does not combine: prescribe where support begins and make M absorb every fixed denominator as the support grows. Equal spacing turns each cancellation equation into a polynomial root condition. Scaling then makes the coefficients integral and the last coefficient equal to one. Let $D, r \geq 2$ and let ℓ be a positive multiple of $2, \ldots, D$. The roots will be $(d!)^{-\ell/d}$. Put

$$i_j = r + j\ell \ (0 \leq j < D), \quad N = r + (D-1)\ell, \quad \alpha_d = (d!)^{\ell/d},$$

$$\prod_{d=2}^{D} (\alpha_d X - 1) = \sum_{j=0}^{D-1} h_j X^j, \qquad A = \prod_{d=2}^{D} \alpha_d.$$

At index i_j put the coefficient

$$c_{i_j} = \frac{N! h_j}{A i_j!},$$

and put every other coefficient equal to zero. We verify that these coefficients are integers and that $V_2 = \cdots = V_D = 0$. For $2 \leq d \leq D$, divisibility of ℓ by d gives $\lfloor i_j/d \rfloor = \lfloor r/d \rfloor + j\ell/d$. Hence

$$V_d(c) = \frac{N!}{A (d!)^{\lfloor r/d \rfloor}} \sum_{j=0}^{D-1} h_j \alpha_d^{-j} = 0,$$

because α_d^{-1} is a root of the displayed polynomial. Evaluating that polynomial at 1 gives M

$$M = N! \prod_{d=2}^{D} (1 - 1/\alpha_d) > 0.$$

For integrality, each term of h_j/A is, up to sign, the reciprocal of a product of $D-1-j$ distinct factors α_d. In that denominator, each selected d contributes ℓ/d blocks of size d. The total size is $(D-1-j)\ell = N - i_j$, so the multinomial coefficient shows that this denominator divides $N!/i_j!$. Thus every c_{i_j} is an integer. Since $h_{D-1} = A$, the coefficient at N is 1, and the coefficients have gcd one. This does not establish minimal support. The smallest permitted step is $\ell = \operatorname{lcm}(2, \dots, D)$; taking $D = r = \ell = 2$ recovers $-6e_2 + e_4$.

The stronger divisibility

$$r! \prod_{d=2}^{D} (\ell/d)! \prod_{d=2}^{D} (\alpha_d - 1) \mid M$$

follows because

$$\frac{N!}{r!\, A \prod_{d=2}^{D} (\ell/d)!}$$

is an integer: it counts partitions into one distinguished block of size r and, for each d, an unordered collection of ℓ/d blocks of size d. Their sizes sum to N. Using $M = (N!/A) \prod_{d=2}^{D} (\alpha_d - 1)$ now gives the asserted divisibility. Put $v = \max(r, \ell/2)$, an integer since $2 \mid \ell$. The displayed factor divides M, so $v! \mid M$. Since $D \le \ell \le 2v$ and $r \le v$,

$$N = r + (D-1)\ell \le 4v^2 - v < 4v^2, \qquad v \ge \lfloor \sqrt{N}/2 \rfloor + 1.$$

Consequently $(\lfloor \sqrt{N}/2 \rfloor + 1)! \mid M$. Every positive integer $q \le \lfloor \sqrt{N}/2 \rfloor + 1$ therefore divides M, as does every divisor of that factorial. In particular, $N \ge 4(q-1)^2$ suffices for $q \mid M$. This includes denominators growing with N.

The same vector also supplies the size estimate needed for a tail test. Every factor $1 - 1/\alpha_d$ lies strictly between 0 and 1, so $0 < M < N!$ and

$$0 < M(S - H_N) < \frac{2M}{(N+1)! - 1} < \frac{2}{N},$$

where $(N+1)! - 1 > NN!$ for $N \ge 2$. Thus the family supplies both eventual divisibility by each fixed q and the size condition $M < (N+1)! - 1$, without an extra hypothesis. It does not yet compare this tail with the finite part's gap to the next integer. Both may shrink together. Before studying that arithmetic question, we show why the cofactor formula gives the same primitive vectors.

1.5 Cofactors give the same primitive vector

A determinant formula may appear to offer a different family. For positive starting index, however, its primitive normalization gives exactly the preceding progression vector, up to sign. We first verify the determinant construction, then identify the two formulas. For integers $n, t \ge 0$, put $s_n = ((n+2)!)^2$ and $i_j = (t+j)s_n$ for $0 \le j \le n+1$. This step

size matches the recorded construction; the determinant argument only needs a positive step divisible by every $d = 2, \dots, n+2$. Here the factorial-weight formula is also allowed at index 0; the case $t = 0$ will show why a lower support condition is necessary. Form the integer matrix A with first row $(i_j!)_j$ and weighted-sum rows $(W_{d,i_j})_j$ for $2 \le d \le n+2$.

To see that $\det A \neq 0$, divide column j by $i_j!$. Since $d \mid s_n$, the dth weighted-sum row becomes $((d!)^{-(t+j)s_n/d})_j$. Removing the nonzero factor $(d!)^{-ts_n/d}$ from that row leaves a Vandermonde matrix with nodes

$$1,\ (2!)^{-s_n/2},\ \dots,\ ((n+2)!)^{-s_n/(n+2)}.$$

These nodes are distinct: $(d!)^{1/d}$ is strictly increasing, since $d! < (d+1)^d$. The Vandermonde determinant is the product of their pairwise differences [32, (1.3.13)], and is therefore nonzero. None of the row or column factors removed is zero, so the original determinant is nonzero too.

Let c be the cofactor vector of the first row, and let N_d be the determinant obtained by replacing that row with $(W_{d,i_j})_j$. Cofactor expansion gives $M(c) = \det A \neq 0$ and $V_d(c) = N_d$. For $2 \le d \le n+2$ the replacement duplicates a row, so $V_d(c) = 0$. By (1) each $(V_d(c) - M(c))/(d! - 1)$ is an integer and vanishes for $d > \max_j i_j$, so

$$\mathcal{R}_{n,t} := \sum_{d>n+2} \frac{N_d}{d! - 1}, \qquad \mathcal{R}_{n,t} - \det(A)\, S \in \mathbb{Z}.$$

A family with $\min_j i_j = t\, s_n \to \infty$ and $\mathcal{R}_{n,t} \notin \mathbb{Z}$ would prove irrationality. Indeed, for a fixed denominator q, every entry of the first row is divisible by q once the least support index is at least q, so $q \mid \det A$. If $S = a/q$, the integer-difference identity would then force $\mathcal{R}_{n,t} \in \mathbb{Z}$. Conversely, if S is irrational, that same identity and $\det A \neq 0$ make every $\mathcal{R}_{n,t}$ irrational. Taking, for example, $n = 0$ and $t \to \infty$ gives the required family. Thus nonintegrality along such a family is equivalent to the original irrationality question, not a weaker conjecture obtained from the construction.

The condition $ts_n \to \infty$ cannot be replaced by unboundedness of (n, t): at $t = 0$, the least supported index stays zero. The coefficient at i_0 really is nonzero: its cofactor uses columns $1, \dots, n+1$, and after nonzero row and column scalings it is a Vandermonde determinant on the weighted-sum nodes. The eventual identity $N_d = \det A$ determines the tail, but not the finite intermediate sum, whose terms may have either sign. Thus the determinant proves cancellation, but its nonzero value does not prove nonintegrality of the remainder.

For $t \ge 1$, primitive normalisation identifies this vector exactly. Take $D = n+2$, $\ell = s_n$ and $r = ts_n$ in the progression construction of §1.4, and put $N = r + (D-1)\ell$. For any coefficient vector supported on these D indices, the equations $V_2 = \cdots = V_D = 0$ say that the polynomial

$$\sum_{j=0}^{D-1} c_{i_j} i_j! X^j$$

vanishes at the $D-1$ distinct points $\alpha_d^{-1} = (d!)^{-\ell/d}$, $2 \le d \le D$. Its degree is at most $D-1$, so it is a scalar multiple of $\prod_{d=2}^{D}(\alpha_d X - 1)$. Thus the rational solution space is

one-dimensional. The progression vector has coefficient 1 at N, so every integer solution is its integer multiple, with multiplier c_N. In particular, the cofactor gcd is $|c_N|$, and the primitive cofactor vector is the progression vector up to sign. Choosing the sign so that $M > 0$ gives

$$N! \prod_{d=2}^{D} (1 - (d!)^{-\ell/d}).$$

For example, $n = 0, t = 1$ gives support $\{4, 8\}$, cofactor vector $2520e_4 - 6e_8$, and primitive vector $-420e_4 + e_8$ with $M > 0$.

The factorial divisibility proved for the progression coefficient M therefore also applies to this primitive vector. Normalisation still leaves the nonintegrality condition equivalent to irrationality along families whose least support index tends to infinity: M is nonzero, every factorial below that index divides it, and (1) gives the integer-difference identity. This identification does not estimate the distance from an integer. The case $t = 0$ is excluded from the identification with the stated progression construction, whose indices are at least 2.

2 Prime powers in reduced denominators

Let p be a prime, and let $v_p(a)$ denote the exponent of p in a positive integer a. We determine which prime powers of L_M survive in $\mathrm{den}(H_M)$. At the largest exponent, this follows by specialising Louwsma and Martino's valuation formula for an elementary symmetric sum [4, Lemma 4.1, p. 10]. Indeed, $\sum_i 1/x_i = (\sum_i \prod_{j \neq i} x_j)/\prod_i x_i$ for positive integers x_i. Their lemma evaluates the numerator's valuation; subtracting the product's valuation gives the reciprocal-sum form. The proof below uses the least common multiple instead of the product.

The distinction is already visible in $1/3 + 1/15 = 2/5$: the common denominator contains 3, but the reduced denominator does not. By contrast, $1/9 + 1/45 = 2/15$ only lowers the exponent of 3 from 2 to 1. The theorem below tests whether the largest exponent is preserved; a failure of that test need not remove the prime completely. The formula is standard. Its mechanism is simple: after clearing by the lcm and reducing modulo p, only denominators carrying the largest power of p contribute. The factorial examples and later tail comparison concern this particular series.

Theorem 2.1 (maximal prime-power survival)**.** *Let $M \geq 2$, let p be a prime dividing L_M, and put $e = v_p(L_M)$. Let $J = \{n : 2 \leq n \leq M,\ v_p(d_n) = e\}$ and write $d_n = p^e u_n$ for $n \in J$. Then, with inverses in $\mathbb{F}_p$,*

$$v_p(\mathrm{den}(H_M)) = e \quad \Longleftrightarrow \quad \sum_{n \in J} u_n^{-1} \neq 0 \quad \text{in } \mathbb{F}_p. \tag{8}$$

Proof. Write $L_M = p^e W$ with $p \nmid W$. If $v_p(d_n) < e$, then $p \mid L_M/d_n$. If $n \in J$, then $u_n(L_M/d_n) = W$, so $L_M/d_n \equiv W u_n^{-1} \pmod{p}$. Summing over $2 \leq n \leq M$ gives

$$A_M \equiv \frac{L_M}{p^e} \sum_{n \in J} u_n^{-1} \pmod{p}. \tag{9}$$

Finally $\operatorname{den}(H_M) = L_M/\gcd(A_M, L_M)$, whose p-valuation is e exactly when $p \nmid A_M$. Since W is invertible modulo p, that is exactly the stated condition. □

The complete valuation identity is

$$v_p(\operatorname{den}(H_M)) = \max\{0, e - v_p(A_M)\}.$$

A zero residue in (8) therefore lowers the exponent, whereas complete cancellation requires $v_p(A_M) \geq e$. For $e > 1$, reduction modulo p alone cannot distinguish partial from complete cancellation. The proof uses no property of factorials beyond the displayed denominators, so it applies to any finite sum of reciprocals of positive integers. For $d_n = n! - 1$, the examples below show that complete cancellation can actually occur; the general valuation formula alone does not predict it.

Two complete cancellations. A unique maximal exponent makes the residue sum a single nonzero term, so the condition holds automatically. With several maximal terms it can fail. The following examples show cancellation in the factorial sequence itself, not just in arbitrary rational sums. Take $M = p - 1$. The table lists every index with $p \mid d_n$, and every one has valuation exactly one.

p	indices n	$d_n/p \pmod p$	inverses modulo p
139	69, 122, 137	6, 49, 73	116, 122, 40
2593	349, 2243, 2591	1508, 1566, 1678	1367, 356, 870

The inverse sums are $278 = 2 \cdot 139$ and 2593. By Theorem 2.1, $139 \nmid \operatorname{den}(H_{138})$ and $2593 \nmid \operatorname{den}(H_{2592})$: each prime divides the common denominator of its partial sum and vanishes from its reduced denominator. The listed indices and their valuations are checked by the recurrence

$$r_1 = 1, \qquad r_n \equiv n r_{n-1} \pmod{p^2}, \qquad 0 \leq r_n < p^2,$$

run through $n = p-1$. At an index with $r_n \equiv 1 \pmod p$, the cofactor is $(r_n - 1)/p$ modulo p. Valuation at least two would give $r_n = 1$, which occurs for neither prime among the tested indices $2 \leq n \leq p - 1$. That enumeration is a finite calculation separate from the theorem.

The recurrence is run modulo p^2, not merely modulo p: reduction modulo p would find the indices with $p \mid n! - 1$, but would not determine whether $p^2 \mid n! - 1$. This is why the zero inverse sums prove complete cancellation in these examples.

For each prime $p \geq 3$, the exponent $v_p(\operatorname{den}(H_M))$ is constant for $M \geq \max(2, p - 2)$. Wilson's theorem gives $(p-1)! - 1 \equiv -2 \pmod p$, and $n! - 1 \equiv -1 \pmod p$ for $n \geq p$. Hence every summand after index $p - 2$ has denominator coprime to p. If a reduced fraction a/b is followed by a summand $1/d$ with $p \nmid d$, their sum has numerator $ad + b$ over bd. When $p \mid b$, this numerator is nonzero modulo p, so the exponent of p in the reduced denominator stays the same. When $p \nmid b$, the new denominator remains coprime to p. Consequently, the two cancellations persist in every later partial sum:

$$139 \nmid \operatorname{den}(H_M) \quad (M \geq 137), \qquad 2593 \nmid \operatorname{den}(H_M) \quad (M \geq 2591).$$

The primes still divide the corresponding common denominators L_M. These are assertions about the partial sums; they do not constrain a hypothetical denominator of their real limit S without a further approximation estimate.

We next ask where a prime first divides a denominator d_m. This means it divides none of $d_2, \dots, d_{m-1}$; it may still divide a later denominator, so uniqueness at its first occurrence gives no uniqueness in every larger block.

Proposition 2.2 (cofinal first prime occurrences)**.** *For every integer $B \geq 0$ there are a prime q and an integer $m > B$ with $m < q$, $q \mid m! - 1$ and $\gcd(q, k! - 1) = 1$ for every k with $2 \leq k < m$.*

Proof. Choose a prime $q \geq B! + 5$. Wilson's theorem gives $(q-2)! \equiv 1 \pmod{q}$, so there is a least index $m \geq 2$ with $q \mid m! - 1$, and $m \leq q - 2$. If $m \leq B$, then $q \leq m! - 1 \leq B! - 1$, contradicting the choice of q. Hence $m > B$, and minimality gives the coprimality assertions. □

The construction proves cofinality of first occurrences, but not the inequality involving q needed for the additional two-factor test in §5. The sufficient tail inequality there does not require choosing a prime q. Wilson reflection limits what can be inferred from a large prime factor alone: for odd $2 \leq n < q$ with $q \mid n! - 1$, the identity $(q - 1 - n)!\, n! \equiv (-1)^{n+1} \pmod{q}$ gives the reflected congruence at $q - n - 1$. It gives a second summand denominator only when $q \neq 2n + 1$ and $n \leq q - 3$. At the Wilson endpoint $n = q - 2$, it instead returns index 1, where $1! - 1 = 0$ is not a denominator of this series. The reflection identity is classical; see Stewart [8, p. 462, (4)].

3 Growth of the common denominator

A large product of denominators gives a large least common multiple only if their overlap is controlled. We use the last k denominators: their logarithmic product contributes about $kN \log N$, while the pairwise-gcd estimate costs about $(k^3/6) \log N$. With $k \sim \alpha\sqrt{N}$, the leading coefficient is $\alpha - \alpha^3/6$, maximized at $\alpha = \sqrt{2}$. The lemmas below make this balance rigorous and yield $\liminf \log L_N/(N^{3/2} \log N) \geq 2\sqrt{2}/3$. Here L_N is the common denominator before reduction, not $\operatorname{den}(H_N)$.

Lemma 3.1 (product, least common multiple, pairwise gcd)**.** *For positive integers $x_1, \dots, x_k$,*

$$\prod_{i=1}^{k} x_i \;\Big|\; \operatorname{lcm}(x_1, \dots, x_k) \prod_{i<j} \gcd(x_i, x_j).$$

Proof. For each prime r, arrange $a_i = v_r(x_i)$ in increasing order. The valuation of the right-hand side is $a_k + \sum_{i=1}^{k-1} (k-i)a_i$, whereas that of the left-hand side is $\sum_{i=1}^{k} a_i$. Their difference is $\sum_{i=1}^{k-1} (k-i-1)a_i \geq 0$. This proves divisibility at every prime. □

The next divisibility is the case $P = -1$ of the relation $\gcd(i! + P(i), j! + P(j)) \mid (j!/i!)P(i) - P(j)$ for $P \in \mathbb{Z}[X]$, obtained by multiplying $i! + P(i)$ by $j!/i!$ and subtracting $j! + P(j)$. Luca and Shparlinski use this relation to bound a common divisor of $n! + P(n)$ and

$(n+h)!+P(n+h)$ in the proof of their Lemma 5 [13], and Lai uses it for a common prime-power divisor [14, proof of Lemma 2.4, display (2.5)].

The earlier spacing method of Erdős and Stewart [18, §3, pp. 516–517] is related background. We now state only the subtraction needed for the lcm argument, rather than importing the prime-factor estimates of those papers.

Lemma 3.2 (factorial-gap gcd)**.** *For $2 \le i < j$, the integer $g = \gcd(i!-1, j!-1)$ divides $j!/i!-1$, and $g \le j!/i!-1 < j^{j-i}$.*

Proof. The identity $(j!/i!)(i!-1)-(j!-1) = 1-j!/i!$ shows that g divides $j!/i!-1$. This integer is positive. Moreover, $j!/i!$ is a product of $j-i$ integers at most j, so $g \le j!/i!-1 < j^{j-i}$. □

Lemma 3.3 (segment inequality)**.** *For $2 \le k \le N-1$,*

$$\sum_{n=N-k+1}^{N} \log(n!-1) \;\le\; \log L_N + \binom{k+1}{3} \log N. \tag{10}$$

Proof. Apply Lemma 3.1 to the k denominators with indices $N-k+1, \dots, N$. Their lcm divides L_N. Lemma 3.2 bounds the logarithm of each pairwise gcd by $(j-i)\log N$. Since

$$\sum_{N-k+1 \le i < j \le N} (j-i) = \sum_{h=1}^{k-1} h(k-h) = \binom{k+1}{3},$$

taking logarithms gives (10). □

Theorem 3.4 (common-denominator growth)**.**

$$\liminf_{N\to\infty} \frac{\log L_N}{N^{3/2}\log N} \;\ge\; \frac{2\sqrt{2}}{3}.$$

Proof. The sum of the logarithms on a terminal block of length k is of order $kN\log N$, while the gcd estimate costs order $k^3 \log N$. We therefore fix $\alpha > 0$ and take $k = \lfloor \alpha\sqrt{N} \rfloor$. For large N this is an admissible block length. If $u = N-k+1$, then $n!-1 \ge n!/2$ and $\log n! \ge n\log n - n$ give

$$\sum_{n=u}^{N} \log(n!-1) \ge k(u\log u - u) - k\log 2.$$

Here $k = \alpha\sqrt{N} + O(1)$ and $u = N + O(\sqrt{N})$, so $ku\log u = \alpha N^{3/2}\log N + O(N\log N)$. The remaining two terms are $O(N^{3/2})$. Also

$$\binom{k+1}{3}\log N = \frac{\alpha^3}{6} N^{3/2}\log N + o(N^{3/2}\log N).$$

After division by $N^{3/2}\log N$, Lemma 3.3 therefore yields a lower limit of at least $\alpha - \alpha^3/6$. This expression is maximised at $\alpha = \sqrt{2}$, giving $2\sqrt{2}/3$. □

The displayed proof of the lcm bound uses only Lemmas 3.1 and 3.2. Theorem 12 of Garaev, Luca and Shparlinski [7, arXiv v1, p. 16] supplies a different input: a uniform $O(N^{2/3})$ bound for the number of solutions of $n! \equiv a \pmod p$, for fixed $a \not\equiv 0 \pmod p$, in an interval of length N contained in $1 \le n < p$. The weaker lcm deduction from it is retained in §B.3, not used here. No exhaustive priority claim is made for the lcm estimate; its precise elementary derivation and the inherited subtraction are stated so that the claim can be assessed without relying on a search conclusion.

Remark (polynomial shifts). Fix $P \in \mathbb{Z}[X] \setminus \{0\}$. By Lemma 3 of Luca and Shparlinski [13], in the form given by Lai [14, Lemma 2.1], there is $n_0 \ge 2$, depending only on P, such that $n! + P(n) > 1$ for all $n \ge n_0$ and $P(n)(n+1)\cdots(n+h) \ne P(n+h)$ for all $n \ge n_0$ and $h \ge 1$. The nonvanishing has a short direct proof. Beyond a cutoff, P has constant nonzero sign and $P(n+1)/P(n) \to 1$, so $0 < P(n+1)/P(n) < n+1$ for every sufficiently large n. Multiplying these inequalities gives

$$0 < \frac{P(n+h)}{P(n)} < \prod_{j=1}^{h}(n+j) \qquad (h \ge 1).$$

Thus the required subtraction is nonzero for every gap h, not merely for each fixed h after a separate cutoff. Increasing n_0 also makes $n! + P(n) > 1$. The proof of Theorem 3.4 then gives

$$\liminf_{N\to\infty} \frac{\log \operatorname{lcm}\{n! + P(n) : n_0 \le n \le N\}}{N^{3/2}\log N} \;\ge\; \frac{2\sqrt{2}}{3}.$$

For $n_0 \le i < j \le N$, multiplying $i! + P(i)$ by $j!/i!$ and subtracting $j! + P(j)$ gives

$$\gcd(i! + P(i),\, j! + P(j)) \,\Big|\, \frac{j!}{i!}P(i) - P(j),$$

which replaces Lemma 3.2. The right-hand side is nonzero by the choice of n_0, and its absolute value is at most $C_P N^{j-i+\deg P}$ for a constant C_P. Over a block of k indices the pairwise loss therefore grows by $O_P(k^2 \log N)$, which is $O_P(N \log N)$ when $k = \lfloor \alpha\sqrt{N} \rfloor$. Since $n! + P(n) \ge n!/2$ for large n, the lower estimate for the sum of logarithms is unchanged. The cutoff is needed, since for $P = -2$ the factor at $n = 2$ vanishes. The hypothesis $P \ne 0$ is needed as well, since for $P = 0$ the least common multiple is $N!$ whatever the cutoff.

A non-polynomial comparison. For $n! + 2^n - 1$, Luca and Shparlinski [19, Lemmas 2.1–2.3, pp. 860–862] eliminate the exponential term using three indices satisfying the congruence; multiplicative order enters the counting. Their valuation-layer identity [19, (3.2), p. 863] records repeated prime-power divisibility. These are related tools, not inputs to Theorem 3.4, and no extension to general non-polynomial perturbations is asserted.

Within the terminal-block estimate just proved, maximising $\alpha - \alpha^3/6$ gives $2\sqrt{2}/3$. This is an optimisation of that one-parameter bound, not an optimality theorem for all block selections or all gcd arguments. The discarded factorial term has size $N^{3/2}$, so a finite normalised ratio need not be close to the limiting lower bound.

Theorem 3.4 concerns L_N, not $\operatorname{den}(H_N)$. Theorem 2.1 and the cancellations at 139 and 2593 show why these cannot be interchanged: a prime present in the common denominator may be absent after reduction. The two examples establish this possibility, not an asymptotic estimate for cancellation.

Even two consecutive denominators obstruct clearing by the full lcm. For $N \geq 3$, the gcd of $(N-1)!-1$ and $N!-1$ divides $N-1$, whereas $(N-1)!-1$ is coprime to $N-1$. Thus those denominators are coprime, and

$$L_N(S - H_N) > \frac{((N-1)!-1)(N!-1)}{(N+1)!-1} \longrightarrow \infty.$$

This argument does not use the stronger asymptotic growth theorem.

Reduction could still help, but only in the opposite direction: a clearing argument needs a sufficiently small reduced denominator. Write $H_N = \widehat{A}_N/Q_N$ in lowest terms, reserving A_N for the numerator over the full lcm. Under $S = a/q$, with $a \in \mathbb{Z}$ and $q \geq 1$, positivity gives

$$qQ_N(S - H_N) = aQ_N - q\widehat{A}_N \in \mathbb{Z}_{>0}.$$

A contradiction requires $qQ_N(S - H_N) < 1$ at some N. A lower bound on Q_N cannot supply this upper estimate. For comparison, the rational limit 1 and the approximants $1 - 1/Q_N$ have arbitrarily large reduced denominators without any irrationality. The inequality $d_N(S - H_N) < 1$ alone would not repair the argument, because d_N generally does not clear the earlier summands of H_N. The prime-power analysis describes which factors survive reduction, but it does not yet supply the required upper control on Q_N.

For linear forms in p-adic zeta values, with $p \geq 5$ a fixed prime and n the index of the form, Lai, Lupu and Sprang quantify a saving of this kind. They bound the denominators of the coefficients by powers of $\operatorname{lcm}(1, \dots, n)$, show that the bound may be divided by a product Φ_n of prime powers, and compute the growth rate of Φ_n [15, Lemmas 5.3–5.7 and 7.3]. That saving enters the inequality [15, (8.1)] under which the rescaled forms, which have integer coefficients, meet the irrationality criterion [15, Lemma 2.1, p. 3] that they quote from Lai [16, Lemma 2.1, p. 4]. Along one unbounded subsequence, the forms must be nonzero and their p-adic absolute values, multiplied by the largest ordinary absolute value of an integer coefficient, must tend to zero. The coefficient bound is archimedean; the bound on the value is p-adic. The conclusion is that at least one of the finitely many p-adic numbers in the forms is irrational, not that each is irrational. The corresponding saving for H_N is the ratio L_N/Q_N.

This ratio is an integer. Merely naming it supplies neither its growth nor a small linear form; the analogy is about what an eventual estimate would have to accomplish. Theorem 2.1 decides, for each prime $p \mid L_N$, whether p divides this ratio; no asymptotic lower bound for the ratio is established here. This paragraph compares the two methods; no p-adic theorem is applied to S.

3.1 Consequences for the support parameter

The lcm estimates also constrain the size of a cancellation vector when its multiplier is bounded by a factorial. At depth $D = 2t^2$, cancellation gives $L_D \mid M$; the next theorem additionally assumes $0 < M < (R+1)! - 1$. Its R is an index, not the remainder or

the modulus R_p of §5. The conclusion bounds the largest supported index only when R denotes that index. Taking R to be a later cutoff gives a bound on the cutoff instead.

Theorem 3.5 (a lower bound for the support parameter)**.** *Let $t, M, R \in \mathbb{N}$ satisfy*

$$t \geq 2^{32}, \qquad M > 0, \qquad L_{2t^2} \mid M, \qquad M < (R+1)! - 1.$$

Then $3t^3 < 2(R+1)$. Consequently no family satisfying these hypotheses for all sufficiently large t has $(R(t)+1)/t^3 \leq 3/2$ eventually, and none has $R(t) = o(t^3)$.

Proof. Put $D = 2t^2$, $k = 2t$ and $u = D - k + 1 = 2t^2 - 2t + 1$. Lemma 3.3 at $N = D$, with $n! - 1 \geq n!/2$ and $\log n! \geq n \log n - n$, and with $L_D \leq M < (R+1)!$, gives

$$2t(u \log u - u - \log 2) < (R+1)\log(R+1) + \binom{2t+1}{3} \log(2t^2). \tag{11}$$

Suppose $R + 1 \leq \frac{3}{2}t^3$. Since $t \geq 16$ one has $\frac{15}{8}t^2 \leq u \leq 2t^2$ and $\log u \geq 2 \log t$, so the left side of (11) is at least $\frac{15}{2}t^3 \log t - 4t^3 - 2t \log 2$. Using $\binom{2t+1}{3} < \frac{4}{3}t^3$ and $\log \frac{3}{2} < \log 2$, the right side is at most

$$\tfrac{3}{2}t^3(\log 2 + 3 \log t) + \tfrac{4}{3}t^3(\log 2 + 2 \log t).$$

Dividing by t^3 and rearranging, (11) would require

$$\tfrac{1}{3} \log t < 4 + \frac{2 \log 2}{t^2} + \tfrac{17}{6} \log 2.$$

But $\log t \geq 32 \log 2$ and $\frac{2}{3} < \log 2 < 1$ give $\frac{1}{3} \log t - \frac{17}{6} \log 2 \geq \frac{47}{6} \log 2 > 5$, while $4 + 2 \log 2/t^2 < 5$. This contradiction proves the finite inequality, and the two sequence conclusions follow. □

The hypotheses are compatible: take $M = L_{2t^2}$ and then choose R large enough. The conclusion rules out subcubic R under the stated size condition; it neither constructs a coefficient vector nor proves irrationality.

Theorem 3.4 raises the asymptotic constant in the same estimate.

Corollary 3.6 (the asymptotic lower bound)**.** *Let $M(t), R(t)$ satisfy $M(t) > 0$, $L_{2t^2} \mid M(t)$ and $M(t) < (R(t)+1)! - 1$ for all sufficiently large t. Then*

$$\liminf_{t \to \infty} \frac{R(t)+1}{t^3} \geq \frac{16}{9}.$$

Proof. Put $r = R(t) + 1$, so $L_{2t^2} \leq M(t) < r!$ and $\log L_{2t^2} < r \log r$. Since

$$\frac{(2t^2)^{3/2} \log(2t^2)}{t^3 \log t} \longrightarrow 4\sqrt{2},$$

Theorem 3.4 gives, for every $\varepsilon > 0$ and all sufficiently large t,

$$\log L_{2t^2} \geq \left(\tfrac{16}{3} - \varepsilon\right) t^3 \log t.$$

Suppose $r \leq ct^3$ for arbitrarily large t, with $0 < c < 16/9$ fixed. On that subsequence $r \log r \leq ct^3(3 \log t + \log c) = (3c + o(1))t^3 \log t$. Choosing $0 < \varepsilon < 16/3 - 3c$ contradicts the lower bound for sufficiently large t. □

This corollary does not assert the strict inequality $R+1 > \frac{16}{9}t^3$ eventually. Indeed, the finite logarithmic constraint (11) allows equality at that constant. To check this directly, set $R+1 = 16t^3/9$ with $t \geq 4$. Since $u = 2t^2-2t+1 \leq 2t^2$ and $x\log x - x$ is increasing for $x \geq 1$, the left side of (11) is at most $4t^3(\log(2t^2)-1) - 2t\log 2$. Subtracting this bound from the right side gives

$$t^3\left(4+\frac{16}{9}\log\frac{16}{9}-\frac{8}{3}\log 2\right)+t\left(\frac{5}{3}\log 2-\frac{2}{3}\log t\right) > \frac{4}{3}t^3-\frac{2}{3}t^2 > 0,$$

where $\log 2 < 1$ and $\log t \leq t$ suffice for the last comparison. Integer pairs with $9(R+1) = 16t^3$ exist for every multiple $t \geq 6$ of 3. This verifies the numerical constraint, not the existence of a coefficient vector with that support. The asymptotic assertion in Corollary 3.6 instead follows from Theorem 3.4.

4 Carries and factorial digits

Instead of clearing the partial sum, we now track the only integer that could equal $m!S$. Under rationality, $m!S$ is integral for all large m; since the positive tail is less than one, it must be the strict successor of $m!H_m$. The recurrence for these successors therefore tests rationality without assuming that $m!$ clears H_m. Put

$$Z_m = \lfloor m!H_m \rfloor + 1, \qquad \Delta_m = Z_{m-1} - (m-1)!H_{m-1} \quad (m \geq 3).$$

Thus $0 < \Delta_m \leq 1$, with $\Delta_m = 1$ when the preceding scaled partial sum is integral. Define the integer carry b_m by

$$Z_m = mZ_{m-1} + 1 - b_m. \tag{12}$$

For example, $2!H_2 = 2$ and $3!H_3 = 36/5$, so $Z_2 = 3$ and $Z_3 = 8$, giving $b_3 = 3\cdot 3+1-8 = 2$. We use $\lfloor x \rfloor + 1$, not $\lceil x \rceil$: at the integral value $2!H_2 = 2$, the required strict successor is 3.

Call $b_m = 1$ a *unit carry*. Put $E_m = m!(S - H_m)$ and $\varepsilon_m = 1/(m!-1)$. Since $d_{n+1} > (n+1)d_n$, a geometric majorant gives the tail estimate

$$0 < E_m < \frac{2\,m!}{(m+1)!-1} < \frac{2}{m} \leq 1 \qquad (m \geq 2). \tag{13}$$

Compare the factorial scaling in Hančl and Tijdeman's tail-integrality lemma for factorial series with integer coefficients [11, Lemma 2.1 and the following remark, p. 385]. Their scaled partial sums are integers; ours need not be, since already $3!H_3 = 36/5$. The next proof instead uses the short positive tail to identify the least integer above a scaled partial sum under a rationality assumption. The carry-defect expansion in §B.2 gives a separate, direct application of factorial-series rationality criteria.

Theorem 4.1 (an exact criterion from successive partial sums)**.** *For $m \geq 3$,*

$$b_m = 1 \iff m \mid Z_m \iff 1+\varepsilon_m < m\Delta_m \leq 2+\varepsilon_m. \tag{14}$$

Moreover

$$S \in \mathbb{Q} \iff b_m = 1 \text{ for all sufficiently large } m, \tag{15}$$

$$S \notin \mathbb{Q} \iff \forall B\ \exists m > B :\ m \nmid Z_m. \tag{16}$$

If $S = a/q$ with $a \in \mathbb{Z}$, $q \geq 1$ and $b_m \neq 1$, then $q \nmid (m-1)!$ and $q \geq m$.

Proof. From $m!H_m = mZ_{m-1} - m\Delta_m + 1 + \varepsilon_m$ one gets $b_m = \lceil m\Delta_m - 1 - \varepsilon_m \rceil$ with $-1 \le b_m \le m-1$, which gives both equivalences in (14), endpoints included.

Suppose $S = a/q$ and $q \mid (m-1)!$. For $j = m-1$ and $j = m$ the number $j!\,S$ is an integer, and $j!H_j = j!\,S - E_j$ with $0 < E_j < 1$ by (13), so $Z_j = j!\,S$. Substituting into (12) forces $b_m = 1$. Every fixed q divides $(m-1)!$ eventually, which gives one direction of (15) and, by contraposition, the final assertion; $q < m$ would imply $q \mid (m-1)!$. Conversely, an eventual unit-carry tail makes $Z_m/m!$ eventually constant, and $0 < Z_m/m! - H_m \le 1/m!$ with $H_m \to S$ identifies that constant as S, which is then rational. Negating (15) gives (16). □

The conclusion $q \nmid (m-1)!$ is about prime-power multiplicities, not just prime factors. It permits all prime factors of q to be at most $m-1$ if one occurs to a larger exponent than in $(m-1)!$.

4.1 The adjacent-unit window condition

Two successive unit carries imply a prime-power size condition after clearing denominators in their defining inequalities. The calculation below shows that the condition follows already from those carries. An additional arithmetic estimate is therefore needed to turn it into a contradiction.

Write $\Delta_k = P_k^\Delta/Q_k^\Delta$ in lowest terms, with $P_k^\Delta, Q_k^\Delta > 0$. In one recurrence step the unreduced denominator is $Q_k^\Delta(k!-1)$; let $G_k = Q_k^\Delta(k!-1)/Q_{k+1}^\Delta$ be the positive integer cancelled in reduction. For $m \ge 3$, put

$$D_m = Q_m^\Delta(m!-1)((m+1)!-1),$$
$$\Omega_m = D_m\left(m(m+1)\Delta_m - m - 2 - \frac{m+1}{m!-1} - \frac{1}{(m+1)!-1}\right).$$

Both are integers. The two unit carries are equivalent to $0 < \Omega_m \le D_m$. Independently of the carries, cancellation in two successive steps gives

$$D_m = Q_{m+2}^\Delta G_{m+1} G_m = Q_m^\Delta(m!-1)((m+1)!-1),$$

and under the pair assumption the offset factors to match, $\Omega_m = P_{m+2}^\Delta G_{m+1} G_m$. Dividing by $G_{m+1}G_m > 0$ reduces the proposed bound to

$$0 < P_{m+2}^\Delta \le Q_{m+2}^\Delta,$$

which every reduced positive gap in $(0,1]$ already satisfies. Thus the resulting gap bound is not an additional restriction on a pair of unit carries. This calculation does not exclude an independent arithmetic obstruction involving the transition numerators or their valuations. Such an obstruction would need information not already implied by the pair assumption; it need not take the form of a bound on the preceding gap Δ_m.

4.2 Factorial digits of $S - e + 2$

The small perturbation from the reciprocal-factorial series can also be isolated term by term. Subtracting $e-2$ leaves the absolutely convergent correction $C = \sum_{n\ge2}(n!(n!-$

$1))^{-1}$. The identity $1/(n!-1) = 1/n! + 1/(n!(n!-1))$ gives

$$S = C + e - 2. \tag{17}$$

The canonical factorial digits of a real x are $a_m(x) = \lfloor m!x \rfloor - m\lfloor (m-1)!x \rfloor \in \{0, \dots, m-1\}$ for $m \geq 2$. At a rational endpoint, the floor convention selects the terminating representation. For example, $1/2 = 1/2!$ also equals $\sum_{m\geq 3}(m-1)/m!$, but its canonical digits are $a_2 = 1$ and $a_m = 0$ for $m \geq 3$, not an eventually maximal tail. With this convention, x is rational exactly when its digits vanish from some index on. The criterion goes back to Cantor [5]; Galambos treats the rationality of Cantor series in [6, Ch. II, §2.1, pp. 21–22], and Koepf and Schmersau prove the irrationality direction for nonterminating factorial expansions whose digits are not eventually maximal [9, Example 3.2, p. 121].

Proposition 4.2 (rationality and factorial residues)**.**

$$S \in \mathbb{Q} \quad \Longleftrightarrow \quad \lfloor m!C \rfloor \equiv -2 \pmod{m} \quad \textit{for all sufficiently large } m.$$

Proof. Put $J_m = \sum_{k=0}^{m} m!/k!$. Then J_m is an integer, every summand with $k < m$ is divisible by m and the summand at $k = m$ is 1, so $J_m \equiv 1 \pmod{m}$; and $0 < m!\,e - J_m < 1$ for $m \geq 2$.

Suppose S is rational. For all large m both $(m-1)!\,S$ and $m!\,S$ are integers, so $m \mid m!\,S$. Multiplying (17) by $m!$ and using $\lfloor -m!\,e \rfloor = -J_m - 1$ gives $\lfloor m!C \rfloor = m!\,S + 2\,m! - J_m - 1 \equiv -2 \pmod{m}$.

Conversely, assume the congruence from some index on. Because $0 \leq a_m(C) < m$, for $m \geq 3$ the congruence is equivalent to $a_m(C) = m - 2$. Choose N beyond the exceptional indices. The canonical expansion gives

$$C = \frac{\lfloor N!C \rfloor}{N!} + \sum_{m>N} \frac{m-2}{m!},$$

and the telescope $\sum_{m>N}(m-1)/m! = 1/N!$ turns this into

$$S = C + e - 2 = \frac{\lfloor N!C \rfloor + 1}{N!} + \sum_{m=2}^{N} \frac{1}{m!},$$

which is rational. □

The proof applies to every real C, not just this series: eventual digits $a_m(C) = m - 2$ characterize rationality of $C+e-2$. Thus $C = 2-e$ satisfies the eventual pattern, whereas $C = 0$ does not. For our particular positive C, neither possibility is decided by a finite digit list.

Negating the eventual congruence gives the equivalent formulation

$$S \notin \mathbb{Q} \quad \Longleftrightarrow \quad (\forall B)(\exists m > B)\ (\lfloor m!C \rfloor + 2) \bmod m \neq 0. \tag{18}$$

It remains to prove failures of this congruence at arbitrarily large indices for this particular C. A finite list of failures does not settle this question, and the equivalence gives no distribution estimate.

Recovering the finite carries. To compare canonical digits with the carries, we must account for the tail removed before taking a floor. Set

$$\delta_m = m! \sum_{n>m} \frac{1}{n!(n!-1)}, \qquad \sigma_m = \mathbf{1}_{\{\{m!C\}<\delta_m\}}.$$

Here $\{x\} = x - \lfloor x \rfloor$. The factorial-tail bound gives $0 < \delta_m < 1/((m+1)! - 1) < 1$. Subtracting δ_m from $m!C$ lowers its floor by one exactly when $\{m!C\} < \delta_m$. Therefore

$$Z_m = m! \sum_{n=2}^{m} \frac{1}{n!} + \lfloor m!C \rfloor + 1 - \sigma_m.$$

Substitution in (12) gives

$$b_m = m - 1 - a_m(C) + \sigma_m - m\sigma_{m-1} \qquad (m \geq 3). \tag{19}$$

When $\{m!C\} = \delta_m$, the scaled partial sum is integral and $\sigma_m = 0$. For a smaller fractional part, subtracting even this small tail crosses an integer. Tail size alone therefore permits neither correction term to be omitted.

4.3 Escape from a smaller interval

Write $\theta_m = \{m!\,S\}$. The next criterion compares a fractional part with the actual tail: θ_{m-1} must leave $[0, E_m/m)$ at arbitrarily large indices. The interval has width less than $2/m^2$. Since E_m contains S, this equivalence is not yet a finite test; the second assertion supplies a sufficient test using the finite quantities Δ_m and ε_m.

Proposition 4.3 (lower-interval criterion)**.**

$$S \notin \mathbb{Q} \quad \Longleftrightarrow \quad \forall B\ \exists m > B:\ E_m \leq m\theta_{m-1}. \tag{20}$$

For $m \geq 3$ the finite condition

$$m\Delta_m \leq 1 + \varepsilon_m \quad \text{or} \quad 1 + \varepsilon_m + \frac{2}{m} \leq m\Delta_m \tag{21}$$

implies the escape inequality in (20). Cofinally many instances of (21) therefore imply $S \notin \mathbb{Q}$.

Proof. If S is rational, the remainders θ_{m-1} vanish eventually while $E_m > 0$, so the escape inequality fails eventually. Conversely, suppose $m\theta_{m-1} < E_m$ for every sufficiently large m. By (13) this gives $m\theta_{m-1} < 1$, so $a_m(S) = 0$ and $\theta_m = m\theta_{m-1}$. Fix N after this recurrence begins. Then $\theta_{N+k} = ((N+k)!/N!)\theta_N < 1$ for every $k \geq 0$, forcing $\theta_N = 0$. Hence $N!S$ is an integer and S is rational.

For the finite implication, suppose $m\theta_{m-1} < E_m$. The tail recurrence is $mE_{m-1} = 1 + \varepsilon_m + E_m$, so $E_{m-1} > E_m/m > \theta_{m-1} \geq 0$, and with $0 < E_{m-1} < 1$ the strict-successor identity gives $\Delta_m = E_{m-1} - \theta_{m-1}$. Hence

$$1 + \varepsilon_m < m\Delta_m = 1 + \varepsilon_m + E_m - m\theta_{m-1} \leq 1 + \varepsilon_m + E_m < 1 + \varepsilon_m + \frac{2}{m},$$

which contradicts (21). □

There are two comparisons to keep separate: a finite sufficient test for escape at one index, and an equivalence when indices are unbounded. The finite test excludes only the open interval $1+\varepsilon_m < m\Delta_m < 1+\varepsilon_m+2/m$; it accepts either endpoint and all values outside. The width $2/m$ refers to $m\Delta_m$; the interval for Δ_m has width $2/m^2$. At one index this is a sufficient test for escape, obtained by replacing E_m with the upper bound $2/m$.

At arbitrarily large integer indices, however, the finite test itself characterises irrationality:

$$S \notin \mathbb{Q} \quad \Longleftrightarrow \quad (21) \text{ holds for arbitrarily large integers } m.$$

Only the forward implication remains to be shown. If S is irrational, Theorem 4.1 gives $b_m \neq 1$ at arbitrarily large indices. By (14), each such index satisfies $m\Delta_m \leq 1+\varepsilon_m$ or $m\Delta_m > 2+\varepsilon_m$. Since $2/m < 1$ for $m \geq 3$, either alternative implies (21). The converse is Proposition 4.3. This combines the two criteria; it does not prove either unbounded occurrence for S. The equivalence is over all integer indices. A restriction to primes gives only a sufficient condition here.

For comparison at a single index, the exact classification is

$$E_m \leq m\theta_{m-1} \quad \Longleftrightarrow \quad b_m \neq 1 \text{ or } a_m(S) = m-1 \qquad (m \geq 3). \tag{22}$$

To verify the classification, use $0 < E_j < 1$ to write

$$Z_j = \lfloor j!S \rfloor + \mathbf{1}_{\{\theta_j \geq E_j\}}.$$

Substitution in the carry recurrence gives

$$b_m = 1 - a_m(S) + m\mathbf{1}_{\{\theta_{m-1} \geq E_{m-1}\}} - \mathbf{1}_{\{\theta_m \geq E_m\}}.$$

Since $0 \leq a_m(S) \leq m-1$, a unit carry has exactly two possibilities: both indicators are zero and $a_m(S) = 0$, or both are one and $a_m(S) = m-1$. In the first case $m\theta_{m-1} = \theta_m < E_m$; in the second, $m\theta_{m-1} \geq m-1 > E_m$. Conversely, $m\theta_{m-1} < E_m < 1$ forces $a_m(S) = 0$ and both indicators to be zero, using $mE_{m-1} = 1+\varepsilon_m+E_m$. It therefore forces a unit carry with zero digit. This proves (22), including the equality case in its left-hand inequality.

Thus escape can occur even when $b_m = 1$, provided the digit is $m-1$. This happens at $m = 52$: exact rational arithmetic gives $b_{52} = 1$ together with $52\,\Delta_{52} > 1+\varepsilon_{52}+2/52$, the margin being about 0.5689674908. This example separates the two events at a single index: the finite test can hold at a unit carry. Their occurrence at arbitrarily large indices nevertheless gives equivalent rationality criteria. Since 52 is composite, the example supplies no prime-index instance.

4.4 The series with denominators $n! + t$

The digit argument also applies to the other shifts mentioned by Erdős. For an integer $t \geq -1$, put $S_t = \sum_{n\geq 2} 1/(n!+t)$ and $C_t = \sum_{n\geq 2} 1/(n!(n!+t))$, so that $S_t = -tC_t + (e-2)$.

The restriction $t \geq -1$ keeps every denominator positive for $n \geq 2$; $t = -2$ is excluded because its first denominator is zero. All series in this identity converge absolutely. The familiar case $t = 0$ is $e-2$; this example will make the residue condition explicit.

Theorem 4.4 (a criterion for the shifts $t \geq -1$)**.** *For every integer $t \geq -1$, the series S_t is rational exactly when*

$$\lceil t\, m!\, C_t \rceil \equiv 2 \pmod{m}$$

for all sufficiently large m, and irrational exactly when that residue is missed cofinally. The member $t = -1$ is S, and the member $t = 0$ is $e - 2$, for which the scaled correction is always 0 and hence misses the residue class at every $m \geq 3$, proving the irrationality of e.

Proof. Put $Y = -tC_t$, so that $S_t = Y + e - 2$. The proof of Proposition 4.2 applies to any real Y: rationality of $Y + e - 2$ forces $\lfloor m!Y \rfloor \equiv -2 \pmod{m}$ eventually. Conversely, that congruence forces the canonical digits of Y to equal $m - 2$ eventually; adding the factorial series of $e-2$ leaves an eventually telescoping tail, hence a rational sum. Finally $\lceil tm!C_t \rceil = -\lfloor m!Y \rfloor$ translates the congruence. Negation gives the cofinal statement. For $t = 0$, the left side is zero, which is not congruent to 2 for $m \geq 3$. □

The $t = 0$ specialisation recovers the classical irrationality of e. For $t \neq 0$, no estimate proving failure of the displayed congruence at arbitrarily large indices is supplied here. This is a statement about the scope of the present argument, not an assertion that every other member has the same current literature status.

5 A complementary-denominator criterion

We seek a multiplier with two properties: it absorbs each fixed rational denominator eventually, and the scaled partial sum has a computable gap to the next integer. Include a factorial to ensure the first property. For the second, the multiplier also absorbs every prime-power level shared by two summand denominators. Any exponent left in the denominator then comes from a unique maximal occurrence, so it cannot cancel on reduction. This multiplier is sufficient, not necessarily minimal. For an integer $p \geq 3$, not necessarily prime, put $I_p = \{2, \dots, 2p-1\}$ and $F_p = (p-1)!$, and define

$$D_p = \operatorname{lcm}_{\substack{i,j \in I_p \\ i<j}} \gcd(d_i, d_j), \qquad C_p = \operatorname{lcm}(F_p, D_p),$$
$$L_p^{\mathrm{blk}} = \operatorname{lcm}(F_p, d_2, \dots, d_{2p-1}), \qquad R_p = L_p^{\mathrm{blk}}/C_p.$$

Prime by prime, D_p records the second-largest exponent, C_p raises it to at least the exponent in F_p, and R_p records what remains below the largest exponent. The factors C_p and R_p need not be coprime: a prime absent from F_p with largest exponent 5 and second-largest exponent 2 occurs to powers 2 and 3 in the two factors. This is an illustration of the valuation calculation, not a claimed pattern in a particular factorial block. It will be useful to retain the ratio $\widetilde{C}_p = C_p/F_p$, so that

$$L_p^{\mathrm{blk}} = C_p R_p = F_p \widetilde{C}_p R_p. \tag{23}$$

Including F_p in both C_p and L_p^{blk} ensures that a fixed rational denominator divides C_p for all large p. The ratio $\widetilde{C}_p$ removes precisely that factorial factor when we estimate the shared part. Put

$$T_p = \sum_{i \in I_p} \frac{L_p^{\mathrm{blk}}}{d_i}, \qquad \rho_p = (-T_p) \bmod R_p, \qquad K_p = 2p^2(2p-1)!,$$

with ρ_p the least nonnegative representative. All of these depend only on a finite partial sum.

For each prime, C_p contains the larger of its exponent in F_p and its second-largest exponent among the denominators. A prime dividing R_p therefore has a unique denominator whose exponent is maximal and exceeds the exponent in F_p, so the argument of Theorem 2.1 applies to it and gives $\gcd(T_p, R_p) = 1$. Thus $R_p = \operatorname{den}(C_p H_{2p-1})$, the reduced denominator of the scaled partial sum, not generally of H_{2p-1}. When $R_p > 1$, this ratio is nonintegral and

$$\frac{\rho_p}{R_p} = \lceil C_p H_{2p-1} \rceil - C_p H_{2p-1} \in (0, 1), \tag{24}$$

the distance to the next integer. When $R_p = 1$, $\rho_p = 0$ but the scaled partial sum is integral, so the least strictly larger integer is at distance 1. The theorem below uses only $R_p > 1$.

At $p = 3$, the denominators $1, 5, 23, 119$ are pairwise coprime. Thus $D_3 = 1$, $C_3 = 2$, $R_3 = 13685$, and

$$C_3 H_5 = \frac{34264}{13685}, \qquad \frac{\rho_3}{R_3} = 3 - \frac{34264}{13685} = \frac{6791}{13685}.$$

The tail estimate in the proof below gives $C_3(S - H_5) < 7/1080 < 6791/13685$. This verifies the comparison for one partial sum. The theorem needs this comparison at arbitrarily large p, so that F_p can absorb any fixed rational denominator.

Theorem 5.1 (a sufficient tail inequality)**.** *Suppose that for every B there is a natural parameter $p \geq 3$ with $p > B$, $R_p > 1$, and*

$$(2p + 1) L_p^{\mathrm{blk}} < K_p \rho_p. \tag{25}$$

Then S is irrational.

Proof. Suppose $S = a/q$ with $q \geq 1$, and choose a parameter in the hypothesis with $p > q$. Since $q \mid F_p \mid C_p$, the number $C_p S$ is an integer strictly above $C_p H_{2p-1}$. The gap identity (24) gives

$$C_p(S - H_{2p-1}) \geq \frac{\rho_p}{R_p}.$$

On the other hand, $1/(n!-1) < 2/n!$ for $n \geq 2p$, and comparison with a geometric series of ratio $1/(2p+1)$ gives

$$S - H_{2p-1} < \frac{2}{(2p)!} \frac{2p+1}{2p} = \frac{2p+1}{K_p}.$$

Multiplication by C_p and (25), with $L_p^{\mathrm{blk}} = C_p R_p$, now give $C_p(S - H_{2p-1}) < \rho_p / R_p$, a contradiction. □

Cancelling R_p in (25) through (23) displays the quantitative issue directly:

$$(2p + 1) C_p < K_p \frac{\rho_p}{R_p}. \tag{26}$$

The factor R_p has cancelled. Thus new prime divisors, even large ones, do not by themselves bound C_p or keep ρ_p/R_p away from zero. Proposition 2.2 supplies the primes but not these two estimates.

Under $R_p > 1$, we have $\rho_p > 0$, and taking logarithms gives

$$\log \widetilde{C}_p - \log \frac{\rho_p}{R_p} < \log \frac{2p^2(2p-1)!}{(2p+1)(p-1)!} = p\log p + (2\log 2 - 1)p + O(\log p).$$

Both terms on the left must be controlled at the same p. The elementary bound $\rho_p \geq 1$ gives only $\rho_p/R_p \geq 1/R_p$, which may be too small. For a concrete comparison, a gap of at least $1/2$ would reduce the sufficient test to $(2p+1)C_p < K_p/2$. At the opposite extreme, the gap $1/R_p$ requires the stronger bound $(2p+1)C_p < K_p/R_p$. Neither type of favourable behaviour is proved here for an unbounded family of these particular prefixes.

A more restrictive test also uses a prime q that first divides $m! - 1$ at an index $m \geq 4$. Choose $p = \lfloor m/2 \rfloor + 1$, which need not be prime. This choice puts m in the block and gives $q \mid R_p$. Indeed, $q > m \geq p$, so $q \nmid F_p$, and no earlier denominator contains q. The only possible later index in the block is $m+1$; when it occurs, $(m+1)! - 1 \equiv m \not\equiv 0 \pmod{q}$. Thus q occurs in just one block denominator and is absent from C_p.

The additional comparison uses the integer R_p/q. Requiring the tail bound to be smaller than both ρ_p and R_p/q gives $\min(\rho_p, R_p/q)$ and is equivalent to the following two inequalities:

$$(2p+1)L_p^{\mathrm{blk}} < K_p \min(\rho_p, R_p/q) \quad \Longleftrightarrow \quad \begin{cases} (2p+1)L_p^{\mathrm{blk}} < K_p\rho_p, \\ (2p+1)C_p q < K_p. \end{cases} \tag{27}$$

The equivalence follows by testing each entry of the minimum and using (23). Both inequalities must hold for the same arbitrarily large parameters. The first alone is already sufficient by Theorem 5.1; the second is an extra restriction, not an equivalent formulation of irrationality.

A fixed small index cannot supply new factors indefinitely. If n is fixed and $p > n! - 1$, then $n! - 1$ divides $(p-1)!$, so none of its prime powers remains outside the factorial base. For example, the factor 719 of $6! - 1$ is useful only over a finite range of p. Any argument using such factors at arbitrarily large p must let their indices increase.

6 Finite denominator exclusions

Two recorded computations give different restrictions on a hypothetical rational denominator. The carry calculation concerns divisibility, and the continued-fraction enclosure concerns size:

$$q \nmid 299999!, \qquad q \geq 2^{39990} > 10^{12038} \tag{28}$$

These restrictions hold simultaneously whenever $S = a/q$, with $a \in \mathbb{Z}$ and $q > 0$; they are not two bounds to compare by taking a maximum. They also hold for an unreduced representation. The size bound passes to multiples of the reduced denominator, as does failure to divide a fixed factorial. The computations were performed outside Lean. The candidate finite-size certificate records that its computation has not been kernel checked, and the full carry certificate has the same outstanding formal obligation.

The two restrictions concern different properties of q. The prime 300007 does not divide 299999!, but lies below the size bound; 299999! exceeds that bound, but fails the nondivisibility condition. A sufficiently high power of a small prime need not divide 299999! either. Thus the carry exclusion is not an exclusion of all denominators with small prime factors. In Sondow's terminology, it says that the least r for which $q \mid r!$ is at least 300000 [21, §3, Theorem 1]. His approximation estimates for e are not being applied to S.

Carry intervals. The GMP calculation encloses each scaled sum by outward-rounded rational intervals, using scale $2^{5025679}$ and 96 guard bits. It uses integer arithmetic throughout. A carry is entered only after the entire interval has been located in one half-open unit cell. All 299998 cells for $3 \le m \le 300000$ were certified in the recorded run. The indices with $b_m = 1$ were exactly

$$52, \quad 591, \quad 1030, \quad 1407, \quad 1438, \quad 2164, \quad 4258, \quad 10991, \quad 21236.$$

In particular $b_{300000} \neq 1$, which gives the first restriction in (28) by Theorem 4.1, and hence also $q \ge 300000$. The independently implemented GMP census is separate from the earlier calculation through 4000. That calculation gives the first six entries of this list and a non-unit carry at its endpoint, so it alone gives $q \ge 4000$. The implication from a non-unit carry to factorial nondivisibility is the recorded kernel-checked result; the full census is not thereby a kernel-checked computation.

A rational enclosure and continued fractions. Set $D = 2^{80000}$ and let N be the first integer satisfying $N! - 1 > D$. Put

$$\ell = \sum_{n=2}^{N-1} \left\lfloor \frac{D}{n! - 1} \right\rfloor, \qquad u = \ell + (N - 2) + \left\lfloor \frac{2D}{N! - 1} \right\rfloor + 1.$$

Each of the $N - 2$ rounded terms loses less than one. Moreover, $(n + 1)! - 1 > (n + 1)(n! - 1)$, so the positive tail from N is less than $(N + 1)/(N(N! - 1)) < 2/(N! - 1)$. These inequalities give the strict enclosure $\ell/D < S < u/D$. The omitted positive tail makes the lower endpoint strict, and the final $+1$ makes the upper endpoint strict even when $2D/(N! - 1)$ is an integer.

The recorded values are $N = 7054$ and $u - \ell = 7053$. Apply Euclid's continued-fraction algorithm simultaneously to the two endpoints, stopping as soon as their integer parts differ or a remainder vanishes. At an inversion, reverse the order of the endpoints. There are 23449 common partial quotients $a_0, \dots, a_{23448}$. Their convergent denominators satisfy

$$Q_{-2} = 1, \qquad Q_{-1} = 0, \qquad Q_j = a_j Q_{j-1} + Q_{j-2}, \qquad Q_{23448} \ge 2^{39990}.$$

Every rational in the open enclosure shares these quotients. Such a rational either equals the last convergent P_{23448}/Q_{23448}, or has a subsequent complete quotient $A/B > 1$, with coprime positive integers A, B, and equals

$$\frac{P_{23448}A + P_{23447}B}{Q_{23448}A + Q_{23447}B}.$$

The determinant identity $P_jQ_{j-1} - P_{j-1}Q_j = \pm 1$ proves that this fraction is reduced and its denominator is at least Q_{23448} [17, §1.12(ii), (1.12.5)–(1.12.7), (1.12.20)–(1.12.21)]. No assumption that the last convergent lies inside the enclosure is needed. The integer inequality $2^{39990} > 10^{12038}$ then gives the decimal bound in (28). The integer program and its recorded output give the 80000-bit scale, width 7053, prefix length 23449, binary exponent 39990 and strict decimal exponent 12038.

Carry computation record. The carry calculation uses the program `scripts/check_erdos68_strict_successor.py` with the GMP backend `scripts/check_erdos68_strict_successor_gmp.cpp`; its output `verification/erdos68-strict-successor.json` records the scale and guard bits above, the unit carries, an event-trace digest and the final enclosure. The SHA-256 digests of the program and the backend, and the payload digest recorded in the output, are

Program: `f25b5bd8ffbfd66bdbb42ca6a07a936aab9b4eb7d3a57446725be55acb642191`.
Backend: `43308ddc3902dd537b789ae6e0648054a0d752adf07e5d6e1d829811e9342083`.
Payload: `044ef27a60f7294c39b8279b7f04e22e1234a942e2862e9e8c64eb66505af29b`.

Both calculations cover finite data. Refining an enclosure can leave the common prefix unchanged, and a later unit carry gives no new carry exclusion. In particular, neither calculation excludes an eventual tail of unit carries. The continued-fraction identities are classical [17, §1.12(ii)]; the numerical parameters in this section are the outputs of the stated computations.

7 Further arithmetic questions

It remains to prove $m \nmid Z_m$ at arbitrarily large indices. Section 4 gives equivalent formulations through the actual remainders in (20) and the finite condition (21), when all integer indices are allowed. The estimates below concern the remaining arithmetic, not further ways to solve the coefficient equations. None establishes the required unbounded set. Restricting indices to primes is an extra condition; selected congruences likewise do not identify the sequence defined by S without its floor relations.

Value sets do not determine the weighted residue sum. Garaev, Luca and Shparlinski's harmonic-sum estimate [20, Theorem 1 and (5)] concerns unconditioned harmonic sums, not sums restricted to the indices where $n! \equiv 1 \pmod q$. The value-set literature addresses different invariants. Banks, Luca, Shparlinski and Stichtenoth [25] provide historical missing-value context; Klurman and Munsch [26, Theorems 2.1–2.2] distinguish unconditional averages from GRH-dependent improvements. Grebennikov, Sagdeev, Semchankau and Vasilevskii [27, Corollaries 1.2 and 1.4] study sizes of value and quotient sets, as do the factorial-residue and representation results of Hu [22] and Garaev and Pardo [23]. None is a noncancellation theorem for the reciprocal unit weights here. A lower bound for the number of distinct residues does give an upper bound on the multiplicity of residue 1, since every other value needs at least one index. But that bound does not determine the reciprocal unit weights or test whether they cancel modulo q. The two explicit cancellations in §2 illustrate why the weights matter. Hu's September preprint [24, Theorem 3.1] considers pairwise distinct maps

$T_\theta(x) = b_\theta + c_\theta/x$ over $\mathbb{F}_p$, with $c_\theta \neq 0$. An internal transition is a pair (x, θ) for which $x \in A \setminus \{0\}$ and $T_\theta(x) \in A$. With at least M distinct such pairs, at most $|A|$ parameters, and a bound $\mu \geq 1$ on the number of ordered parameter pairs representing each nonidentity map $T_\eta T_\theta^{-1}$, the theorem gives $|A| \gg \min\{M, p\}^{8/15}\mu^{-4/15}$. Its incidence estimate comes from Stevens and de Zeeuw [28, Theorem 4]. To apply this to the residue classes of indices satisfying $n! \equiv 1 \pmod q$, one would first have to specify the set A and the transition maps. The number of such indices is not automatically a lower bound for M: different indices may produce the same pair (x, θ), and their factorial values are all 1. One must prove that enough distinct transitions remain, as well as bound the ordered-pair multiplicity μ. The unconditioned value-set theorem does not provide these additional facts.

The shared denominator and the gap must be controlled together. For the blocks selected from first prime occurrences, the second inequality in (27) is

$$\log \widetilde{C}_p < \log\left(\frac{2p^2}{2p+1} \cdot \frac{\prod_{j=p}^{2p-1} j}{q}\right). \tag{29}$$

The first inequality is (25). Both must hold at the same arbitrarily large parameters.

The order of quantifiers is part of the proposed result: for every bound, one parameter must satisfy both inequalities. Two unrelated infinite subsequences would not prove the combined criterion.

For a prime r and an integer $e \geq 1$, let $h_{r,e}(p) = \#\{i \in I_p : r^e \mid i! - 1\}$. The exponent of r in the shared part is at least e exactly when $h_{r,e}(p) > 1$. Removing the exponent already present in F_p therefore gives

$$v_r(\widetilde{C}_p) = \#\{e \geq 1 : h_{r,e+v_r(F_p)}(p) > 1\}.$$

When $r \nmid F_p$, this counts the positive exponents for which two or more denominators are divisible by r^e. The spacing bound $h_{r,e}(p)(e+1) \leq 2p + e - 2$ follows from the gap-product argument: if $i < j$ are two such indices, then $j < r$ and $0 < r^e \leq j!/i! - 1 < r^{j-i}$, so $j - i \geq e + 1$. All indices lie between 2 and $2p-1$, giving the bound when $h_{r,e}(p) > 0$. It also holds when $h_{r,e}(p) = 0$. To estimate $\log \widetilde{C}_p$, these exponent counts must be multiplied by $\log r$ and summed over primes. A count ignoring the sizes of the primes does not give (29). For a prime q, Wilson reflection of an odd index $n < q$ contributes a repeated divisor only when the reflected index is distinct and in the same block, with both indices at least 2. The endpoint $n = q - 2$ reflects to 1 and contributes no such pair, as shown in §2.

Even for two admissible indices, the reflection identity is only modulo q [8, (4), p. 462]. For example, exact multiplication gives

$$609! \equiv 1 \pmod{971^2}, \qquad 361! \equiv 736019 = 1 + 758 \cdot 971 \pmod{971^2},$$

where $361 = 971 - 609 - 1$. Both indices lie in I_{306}, but their factorial denominators have 971-exponents at least 2 and exactly 1, respectively. Thus both indices satisfy the congruence modulo q, but need not satisfy it modulo q^e for $e > 1$. Each level in the

valuation sum requires its own congruence; neither distinctness nor a higher exponent may be inferred from the prime-level identity alone. Theorem 12 of Garaev, Luca and Shparlinski [7, arXiv v1, Thm. 12, p. 16] is a multiplicity bound and supplies no lower bound for the distance to the next integer in (25).

For the first inequality alone, a joint mean bound

$$\frac{1}{|I_X|}\sum_{p\in I_X}\left(\log \widetilde{C}_p - \log\frac{\rho_p}{R_p}\right) < (1-\varepsilon)X\log X$$

on nonempty sets $I_X \subseteq [X,2X]\cap\mathbb{N}$, with $R_p > 1$, a fixed $0 < \varepsilon < 1$, and arbitrarily large X, would suffice: at least one summand is at most the average, while the required upper bound is $\log(K_p/((2p+1)F_p)) = p\log p + O(p)$, uniformly for $p \in [X,2X]$. Two separate estimates on unrelated sets of indices would not give this joint comparison.

What two gap-product tests do not show. For integers $h \geq 1$ and $x \geq 0$, let $F_h(X) = \prod_{j=1}^{h}(X+j)$. Suppose $x+h < q$ for a prime q. If $x!$ and $(x+h)!$ both equal 1 modulo q and the harmonic sums through these indices are equal modulo q, then $F_h(x) = 1$ and $F_h'(x) = F_h(x)\sum_{j=1}^{h}(x+j)^{-1} = 0$ in that field. An exact calculation gives

$$\gcd(F_{12}-1, F_{12}') = (X-4626)(X-7848) \quad \text{in } \mathbb{F}_{12487}[X].$$

Thus a uniform bound of one repeated root is false. But $4626! \equiv 442$ and $7848! \equiv 6300$ modulo 12487, so neither root also satisfies $x! \equiv 1$. Counting repeated roots of $F_h - 1$ therefore does not count the indices required by the factorial problem. These are finite modular calculations, not a bound on the number of indices satisfying both congruences.

Stewart's gap-product estimate [8, Lemma 2, (21)–(24)] requires the two products to be unequal. He supplies that step separately: in the proof of his bound (14), a prime-distribution argument separates the products when the common divisor is large; the smaller-divisor case is immediate [8, p. 467, proof of Lemma 2].

The non-power theorem gives a different argument when the gaps are adjacent. For integers $2 \leq i < j < k$, equality $j!/i! = k!/j!$ would make $k!/i!$ a square, contrary to that theorem [31, Theorem 1]. Selecting two short gaps with nearly equal lengths need not preserve adjacency, so this observation alone cannot replace Stewart's nonvanishing argument. The wider equal-product problem has a separate history [29]; the doubled-length theorem discussed by Nguyen Xuan Tho [30, Theorem 1] is another special case, not arbitrary near-equal-length nonvanishing. No stronger lcm exponent is inferred from these local observations.

Prime powers already seen at earlier indices. Recall that $\Delta_n = P_n^{\Delta}/Q_n^{\Delta}$ is the positive gap in lowest terms. Define

$$\mathcal{A}_n = \prod_{\substack{r \text{ prime},\ r\mid n!-1\\ r\mid k!-1 \text{ for some } 2\leq k<n\\ v_r(Q_n^{\Delta})<v_r(n!-1)}} r^{\,v_r(n!-1)}.$$

Thus the product includes the full power of a prime in $n! - 1$ when that prime has appeared before but has smaller exponent in Q_n^Δ. The recurrence for the gaps is

$$\Delta_{n+1} = n\Delta_n - b_n - \frac{1}{n! - 1}.$$

For a prime r in the product, put $e = v_r(n! - 1)$. The reduced denominator of $n\Delta_n - b_n$ has r-exponent at most $v_r(Q_n^\Delta) < e$, whereas $1/(n! - 1)$ has exponent e. Clearing denominators leaves exactly one numerator term nonzero modulo r, so the reduced denominator of Δ_{n+1} has exponent exactly e. Hence $\mathcal{A}_n \mid Q_{n+1}^\Delta$ and, when $\mathcal{A}_n > 1$, $P_{n+1}^\Delta \not\equiv 0 \pmod{\mathcal{A}_n}$. The earlier occurrence of r is not used in this divisibility argument; it restricts the product to the primes proposed for this approach. A sufficient quantitative estimate, at arbitrarily large indices n, with $m = n + 1$ and $d = \mathcal{A}_n > 1$, is

$$((m + 2)m! - 2)Q_m^\Delta \le m^2(m! - 1)\ (P_m^\Delta \bmod d). \tag{30}$$

It implies the upper alternative of (21). A nonzero least representative can equal one even when the modulus is large. Neither a lower bound for $\log \mathcal{A}_n$ nor infinitely many increases in a previously occurring prime's exponent therefore proves (30): the least representative itself must satisfy the displayed inequality. Repeated prime squares can first occur before the Wilson index $r-2$: for example, 971 divides $361! - 1$ and its square first divides $609! - 1$, with $609 < 971 - 2$. Such an example rules out confinement to that last index but supplies no unbounded family satisfying the required inequality.

The upper alternative in the finite test. It would suffice for (21) to hold at arbitrarily large indices; an unbounded set of prime indices would suffice as well. The upper alternative is exactly $((m+2)m! - 2)Q_m^\Delta \le m^2(m! - 1)P_m^\Delta$. Replacing P_m^Δ by its least nonnegative residue modulo a specified divisor of Q_m^Δ gives a stronger sufficient test. No bound forcing this inequality at arbitrarily large indices is established here. The full recurrence for the actual scaled partial sums, including its initial value, is fixed; a rational example for a weakened recurrence does not satisfy that full specification (§B.7). Nor does a zero canonical digit assert membership in the specified real interval.

Conditions at twice a prime. For an odd prime p, the carry range and $Z_{2p} = 2pZ_{2p-1} + 1 - b_{2p}$ give

$$p^2 \mid Z_{2p} \iff \begin{cases} b_{2p} = 1 \text{ and } p \mid Z_{2p-1}, \text{ or} \\ b_{2p} = 1 + p \text{ and } p \mid 2Z_{2p-1} - 1. \end{cases} \tag{31}$$

Reduction modulo p forces $b_{2p} \equiv 1 \pmod{p}$, and the two displayed values are the only possibilities in $[-1, 2p - 1]$; division by p gives the predecessor conditions. For rational $S = a/q$ and $p > q$, the tail estimate gives $Z_{2p} = (2p)!\,S$, which is divisible by p^2. A proof through $p^2 \nmid Z_{2p}$ must therefore exclude both displayed alternatives at the same prime, for arbitrarily large primes. That is a requirement of this particular test, not of every irrationality argument. Rationality also forces $b_{2p} = 1$ and $Z_{2p-1} = (2p - 1)!a/q$, with $p \mid Z_{2p-1}$: here $q \mid (p-1)!$, and the remaining factorial factors include p. Thus either $b_{2p} \ne 1$ at arbitrarily large odd primes or $p \nmid Z_{2p-1}$ at arbitrarily large odd primes would already contradict rationality. The first is a special case of Theorem 4.1; the second uses the same factorial-clearing argument. Neither unbounded occurrence is proved here.

Criteria from the literature that do not apply. Duverney's Theorem 3.1 assumes, among other conditions, quadratic growth $cu_n^2 \le u_{n+1} \le c'u_n^2$ for positive constants c, c'. For $u_n = n! - 1$, however, $u_{n+1}/u_n^2 \to 0$. His Corollary 3.2 assumes convergence of the signed series $\sum_n (u_{n+1}/u_n^2 - 1)$ in (3.6). Here its terms tend to -1, so that hypothesis fails as well [10, pp. 275, 285–287]. For a single denominator sequence the rapid-growth criterion goes back to Erdős [3, Theorem 1, p. 1]; Barreto, Kang, Kim, Kovač and Zhang treat products of consecutive denominators and weighted extensions [12, Thms. 2–3 and Rem. 4]. For $a_n = n! - 1$, $\log(n! - 1) = O(n \log n)$ makes $(n! - 1)^{1/\psi^n} \to 1$ for every fixed $\psi > 1$, so neither Erdős's hypothesis $\limsup_n a_n^{1/2^n} = \infty$ nor the growth hypotheses of their Theorems 2 and 3 hold. These criteria do not decide S; the useful content is the target these papers identify: a subsequence with clearing integers small enough for the corresponding remainders, or enough exact cancellation in those integers. Theorem 3.4 measures the cost of clearing all summands before reduction; it does not bound cancellation in their sum. Theorem 2.1 tests whether a maximal prime power survives reduction. Neither supplies the upper bound for a clearing integer, relative to its remainder, that these arguments need.

No irrationality conclusion is obtained here. The finite conclusion is that every rational representation $S = a/q$, $q > 0$, must satisfy both $q \nmid 299999!$ and $q \ge 2^{39990} > 10^{12038}$. The criteria above require a condition at arbitrarily large indices. Extending either finite computation alone does not establish it.

Sources and evidence

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The verification record gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded.

The margin marks and the table below identify the recorded Lean statements and supporting formulas at their original revisions. A link to a nearby statement does not certify a further deduction in the prose. The two partial-sum cancellations, the index-52 example and the continued-fraction enclosure use the finite integer calculations described above. The carry computation through 300000 uses a separate exact-interval computation outside Lean, described in §6.

Statement	Lean statements of its steps
Theorem 2.1	maximal-power survival, with the residue formula and the denominator form
Proposition 2.2	first prime occurrences at arbitrarily large indices; the reflected divisibility

Statement	Lean statements of its steps (continued)
Theorem 4.1	failures of divisibility for the next integer; denominator exclusions: the size bound and the factorial nondivisibility
Proposition 4.2	factorial residues and rationality
Proposition 4.3	the equivalent fractional-part inequality; the finite window, the irrationality implication, the classification (22), the radius bound
Theorem 4.4	the criterion for the shifted series; the escape form, the member $t = -1$, and the irrationality of e
Theorem 5.1	the prime-parameter specialization; the margin mark gives the natural-parameter statement; the coprimality (24)
Equation (27)	the two simultaneous inequalities, the floor identity, the irrationality implication
Removal of a fixed denominator factor	a fixed factor eventually divides the factorial
Weight identities used in Theorem 1.1 and (1)	integrality of the weights, the denominator-times-weight identity
Theorem 3.5	finite radius bound; the sequence form, the little-o form
Changing one weighted sum V_p	adjustment at sufficiently large indices, the residual identity
Equation (31)	doubled-prime criterion
Canonical factorial digits	termination equivalence
Amplification modulus	divisibility, nonvanishing

Attribution. Wilson's theorem and the Wilson reflection identity are classical, the latter recorded by Stewart [8, p. 462, (4)]. The factorial-digit termination criterion goes back to Cantor [5]; Koepf and Schmersau prove its irrationality direction for digits that are not eventually maximal [9, Example 3.2, p. 121], and Galambos treats rationality criteria for Cantor series [6, Ch. II, §2.1]. Here the identity $C = S - e + 2$ gives the eventual digit value $m - 2$ and extends to the shifted family. The comparison with Hančl and Tijdeman [11, Lemma 2.1 and the following remark, p. 385] concerns factorial scaling: their lemma makes a scaled tail integral, whereas the proof in §4 identifies the next integer above a generally nonintegral scaled partial sum. The multiplicity bound is Garaev, Luca and Shparlinski's [7], and the lcm deduction from it is not theirs. The divisibility in Lemma 3.2 is the case $P = -1$ of the relation used in the proof of Lemma 5 of Luca and Shparlinski [13] and at display (2.5) of Lai [14]. The nonvanishing cutoff for polynomial shifts is Lemma 3 of [13], restated with the bound $n! + P(n) > 1$ in [14, Lemma 2.1]. The survival criterion is a specialisation of Louwsma and Martino's valuation formula [4, Lemma 4.1, p. 10]. The single-denominator growth criterion is Erdős's

[3, Theorem 1, p. 1], and the continued-fraction identities are those of [17, §1.12(ii)]. The deductions from Wilson's theorem, the conditional finite support bound and Theorem 3.4 are proved above. No further priority claim is inferred from this comparison.

A Guide to the formal sources

This concordance lists the floor, support and divisibility identities used alongside the principal results. All links retain their original source revisions. The last subsection contains the historical links from the short paper, with their targets unchanged.

- the factorial-scaled floor of a rational number
- eventual zero digits of a rational number
- a digit as the floor of a scaled fractional part
- the digit is smaller than its radix
- the recurrence for fractional parts
- the finite factorial expansion with its remainder
- a zero remainder forces all later digits to vanish
- eventual zero digits force a zero remainder
- a zero factorial remainder implies rationality
- the scaled partial sum and limit select the same integer
- non-unit carries at arbitrarily large indices imply irrationality
- every rational denominator is at least sixty-seven
- the exact interval condition for the lower endpoint
- a prime-product bound gives a late first occurrence
- a reflected prime survives normalisation under the block and upper-half hypotheses
- two distinct in-block indices under the same reflection hypotheses
- the selected prime powers divide the remaining quotient
- the factors one and a selected prime in the chosen block
- the exact integer interval for two consecutive unit carries
- the two denominator-reduction factors telescope
- the denominator after two steps
- the offset equals the later numerator times the reduction factors
- factorisation when the floor in the weights is constant
- vanishing of the value of M on one such interval

- factorisation on the interval from d to twice d
- a nonzero value of M forces an index at least twice d
- a cubic lower bound for the support parameter
- exclusion of the stated eventual cubic upper bound
- no change unless the divisor divides the index
- the strict rational bounds for the example remainder
- the consecutive-term quadratic ratio tends to zero
- divergence of the absolute quadratic-ratio errors
- the iterated-root growth expression tends to one
- updating the pairwise gcd lcm after insertion
- updating the lcm with a fixed prescribed factor
- exact removal of the common part with the base
- the valuation after removal of the base
- the shared part times the lcm divides the product
- fixed denominator indices are eventually absorbed by the factorial
- the shared part is bounded by the product divided by the lcm
- the surviving valuation after removal of the factorial base
- a shared prime power is bounded by the gap product
- two indices sharing a prime power are farther apart than its exponent
- the resulting bound on the number of such indices
- the total relevant exponent is less than the prime
- the total relevant exponent is less than the block diameter
- a surviving prime has square greater than the parameter minus one
- the projection moduli have the required lcm
- the two-factor bound is no larger than the full residue bound
- the valuation counts exponents occurring at two indices
- a finite exponent range suffices for that count
- the exact repeated-power criterion above the factorial base

A.1 Source links from the short paper

These links collect the declarations formerly listed in the short paper's source appendix, at the same historical revisions and paths. They supplement the concordance above. No fresh compilation or independent review is asserted by a link, and a theorem about a conditional family leaves its hypotheses to be established.

- maximal prime power survival.
- the explicit vector U_9.
- its weighted sums.
- divisibility of $V_d(\lambda) - M(\lambda)$ by $d! - 1$.
- the Lean file on the integer basis.
- an account of the integer basis.
- an account of the primitive vector.
- the finite formula for the gcd.
- the finite gcd verification script.
- a script.
- Generic gcd lemmas in the Lean sources.
- the liminf bound for the common denominator.
- the inequality for a terminal block.
- the vector $2e_3 - e_4$.
- the exact floor correction for the companion series.
- the prime-power noncancellation criterion.

The remaining links from the short paper use revision `99f4bf47422a`; the links in the preceding concordance use this record's own revision. The following numbers link to source lines within each named module, retaining both revisions where they differ.

`AdjacentUnitCarryWindow.lean`
148, 215, 243, 337.

`ChannelBreakpointRigidity.lean`
71, 91, 101, 130.

`ChannelIntegralCongruence.lean`
868, 885, 905, 1084, 1103.

`DivisorFactorialCentre.lean`
34, 39, 44, 49, 80, 88, 102, 129, 146, 170, 179, 188.

`EndpointWeightedPrivateSupport.lean`
4766.

`FactorialChannelCertificate.lean`
25, 39, 43, 51, 55, 59, 65, 105, 109, 112, 119, 126, 133, 140, 151, 156.

`FactorialZeroPlateau.lean`
876, 940.

`PrimeUnitTranslator.lean`
1559.

`PrimeZeroBranch.lean`
6099.

The hypotheses about short intervals, least prime factors, and the nonzero real remainders are stated in the mathematical sections above. They remain hypotheses when a declaration link is followed.

Earlier links for the consolidated criteria. The four criteria treated here retain the short paper's earlier source and comparison links below. These identify historical declarations and recorded comparisons; they are not new executions.

- The constant-floor support statement: the recorded declaration list, the comparison record.
- The carry characterisation: the earlier non-unit-carry implication, the next-integer criterion, the recorded combined statement, the comparison record.
- The complementary-denominator criterion: the comparison record.
- The companion-digit criterion: the recorded formal statement, the comparison record.

B Further deductions and limitations

The following sections give additional digit identities, denominator calculations and finite examples. For each unsuccessful argument we specify the missing estimate or the counterexample to its inference. The resulting limitations apply to those arguments, without ruling out other uses of the same identities.

B.1 Factorial digits and their remainders

We derive the factorial-digit identities used in §4 directly from their remainders. For a real x, start with $\theta_0 = \{x\}$ and, for $m \geq 1$, put

$$a_m = \lfloor m\,\theta_{m-1} \rfloor, \qquad \theta_m = m\,\theta_{m-1} - a_m,$$

so that $\theta_m = \{m!\,x\}$ for $m \geq 1$. The recorded formal identities give the floor formula and the digit bounds $0 \leq a_m < m$, the recurrence $\theta_{m+1} = (m+1)\theta_m - a_{m+1}$, the finite telescoping expansion

$$x = \lfloor x \rfloor + \sum_{m=2}^{N} \frac{a_m}{m!} + \frac{\theta_N}{N!},$$

and the rule that a zero remainder at one index forces every later digit to vanish. The rational direction is also checked: if $a \in \mathbb{Z}$, $q \in \mathbb{N}$, and $0 < q \leq n$, then $\lfloor n!a/q \rfloor = (n!/q)a$ and the canonical digit at radix $n+1$ vanishes, so every rational input has an eventually zero expansion. The converse holds for every real input, and gives Cantor's termination criterion [5]

$$x \in \mathbb{Q} \quad \Longleftrightarrow \quad a_m(x) = 0 \text{ for all large } m.$$

If all digits after index $N \geq 1$ vanish, the recurrence gives $\theta_{N+k} \geq (k+1)\theta_N$; since every remainder is below one, $\theta_N = 0$ and $x = \lfloor N!x \rfloor / N!$.

For $x = S$, a zero canonical digit means $m\theta_{m-1} < 1$. The stronger condition $m\theta_{m-1} < E_m$, where $E_m = m!(S - H_m)$, is the lower-interval event used in §4. The reported interval data contain zero digits at $m = 5$ and $m = 23$, but no occurrence of that stronger condition through $m = 100000$.

There is also a useful identity for an abstract rational sequence F_m and integer sequence k_m. Suppose $F_m = mF_{m-1} + 1 + \varepsilon_m - k_m$, and define

$$\delta_m = \lceil F_m \rceil - F_m, \qquad q_m = m\lceil F_{m-1} \rceil + 1 - k_m - \lceil F_m \rceil.$$

Then $0 \leq \delta_m < 1$, and the exact identities are

$$q_m = \lfloor m\delta_{m-1} - \varepsilon_m \rfloor, \qquad \delta_m = m\delta_{m-1} - \varepsilon_m - q_m.$$

The Lean sources prove these identities under the displayed recurrence. They do apply to the actual partial sums: take $F_m = m!H_m$, $k_m = 0$ and $\varepsilon_m = 1/(m! - 1)$ for $m \geq 3$. The recurrence then follows from $H_m = H_{m-1} + 1/(m! - 1)$, starting with $F_2 = 2$. The ceiling code q_m need not equal the carry b_m, because Z_m is the least integer strictly above F_m, not always its ceiling. Precisely, $Z_m = \lceil F_m \rceil + \mathbf{1}_{\{F_m \in \mathbb{Z}\}}$, so

$$b_m = q_m + m\mathbf{1}_{\{F_{m-1} \in \mathbb{Z}\}} - \mathbf{1}_{\{F_m \in \mathbb{Z}\}}.$$

For example, $F_2 = 2$ and $F_3 = 36/5$ give $q_3 = -1$ but $b_3 = 2$. The codes agree whenever both scaled prefixes are nonintegral. Once $F_2 = 2$ and $\varepsilon_m = 1/(m! - 1)$ are fixed, the recurrence with $k_m = 0$ determines every F_m uniquely by induction. Thus matching the recurrence and its initial value is stronger than satisfying selected congruences or carry bounds. This specialization is an ordinary deduction from the partial-sum recurrence, not a new Lean check. It does not show that the rational gaps admit a finite-state description or establish the unbounded non-unit carries needed for irrationality.

B.2 How the classical criteria apply here

Let $s_n < a$ be partial sums tending to a. Koepf and Schmersau show that $\lfloor ns_n \rfloor = \lfloor na \rfloor$ for all sufficiently large n forces a to be irrational [9, Thm. 1.1, p. 117]. The strict inequality matters: if $a = A/B$ were rational, then every sufficiently large multiple n of B would give $\lfloor ns_n \rfloor < na = \lfloor na \rfloor$. Their rational-term version uses a positive integer multiplier p_n and obtains the floor equality from $np_n s_n \in \mathbb{N}_0$ and the strict tail bound $a - s_n < 1/(np_n)$ [9, (2.1) and (2.3), p. 118; Thms. 2.2–2.3, pp. 119–120]. Indeed, integrality makes ns_n a multiple of $1/p_n$, so its gap to the least strictly larger integer is at least $1/p_n$; the increase $n(a - s_n) < 1/p_n$ cannot cross that integer. For the choice they record after (2.1), the least common multiple of the reduced summand denominators, here $p_n = \operatorname{lcm}\{k! - 1 : 2 \leq k \leq n\}$, the last two denominators already obstruct the tail bound. As proved in §3, they are coprime for $n \geq 3$, so

$$p_n \geq (n! - 1)((n-1)! - 1).$$

For $n \geq 4$ this gives $np_n > (n+1)! - 1$. Hence the first omitted summand $1/((n+1)! - 1)$ already exceeds $1/(np_n)$, and this p_n cannot satisfy their tail hypothesis. This rules out

the displayed choice of p_n, not the criterion itself. In fact, the least positive integer p_n for which np_nH_n is integral is exactly

$$p_n = \operatorname{den}(nH_n) = \frac{\operatorname{den}(H_n)}{\gcd(n, \operatorname{den}(H_n))}.$$

Indeed, writing H_n in lowest terms shows that its denominator must divide np_n. With this least choice the required inequality is

$$0 < S - H_n < \frac{\gcd(n, \operatorname{den}(H_n))}{n \operatorname{den}(H_n)}.$$

Thus reduction of the partial sum, and the common factor of its reduced denominator with n, determine the best scale available in this criterion. The eventual inequality above remains unproved here; failure of a larger clearing multiplier does not establish its failure.

Duverney's Theorem 3.1 includes the quadratic growth assumption $cu_n^2 \le u_{n+1} \le c'u_n^2$ for positive constants c, c', while $u_{n+1}/u_n^2 \to 0$ for $u_n = n! - 1$ [10, (1.3), p. 275; Thm. 3.1, pp. 285–286]. His Corollary 3.2, which allows signs $a_n \in \{-1, 1\}$, assumes convergence of the signed series $\sum_n (u_{n+1}/u_n^2 - 1)$ in (3.6) [10, Cor. 3.2, p. 287]. Its terms tend to -1 here, so the series does not converge. Lean checks the ratio limit and also the divergence of the absolute-value series; the signed condition fails already by the term test. To see the ratio limit directly, divide numerator and denominator by $(n!)^2$: the numerator tends to zero and the denominator $(1 - 1/n!)^2$ tends to one.

For a strictly increasing sequence (n_k) of positive integers, Erdős proved irrationality of $\sum_k 1/n_k$ under the following conditions, with a fixed $\varepsilon > 0$ [3, Theorem 1, p. 1]:

$$\limsup_{k\to\infty} n_k^{1/2^k} = \infty, \qquad n_k > k^{1+\varepsilon} \quad \text{eventually.}$$

Barreto, Kang, Kim, Kovač and Zhang treat products of consecutive denominators and weighted extensions [12, Thms. 2–3 and Rem. 4, pp. 2–5]. For $a_n = n! - 1$ and every fixed $\psi > 1$,

$$0 \le \frac{\log(n! - 1)}{\psi^n} \le \frac{n^2}{\psi^n} \longrightarrow 0, \qquad (n! - 1)^{1/\psi^n} \longrightarrow 1.$$

Thus neither Erdős's limsup hypothesis, which uses $\psi = 2$, nor the growth hypotheses of their Theorems 2 and 3 hold. The Lean proof of this limit treats the special case $\psi = 2$; the estimate above also covers the other bases greater than one used in those theorems. Their proof of Theorem 3 uses the classical criterion, which they trace to Fourier's proof that e is irrational, that a rational sum of nonnegative rationals with infinitely many positive terms admits no integers D_N clearing the denominators of the partial sums with $\liminf_N D_N r_N = 0$, where r_N is the tail after N terms [12, Lem. 8, p. 6]; their Proposition 12 produces such integers under the hypotheses of that theorem [12, pp. 9–12]. For the present series, even the full summand lcm L_N makes $L_N(S - H_N)$ tend to infinity, as shown in §3. A smaller integer cannot clear every summand. It can nevertheless clear the partial sum after addition: the least positive such integer is $\operatorname{den}(H_N)$, and every other one is a multiple of it. Thus this criterion would require

information about cancellation in the reduced partial sum, not merely an improvement from the product to the lcm. More precisely, if $S = a/q$, then

$$q \operatorname{den}(H_N)(S - H_N) \in \mathbb{Z}_{>0}, \qquad \operatorname{den}(H_N)(S - H_N) \geq \frac{1}{q}.$$

Thus $\liminf_N \operatorname{den}(H_N)(S - H_N) = 0$ is a sufficient target for this rationality criterion; the preceding common-denominator estimates do not establish it.

Dividing the recurrence $Z_m = mZ_{m-1} + 1 - b_m$ by $m!$ and telescoping gives the exact finite identity

$$\frac{Z_M}{M!} = \frac{Z_2}{2!} + \sum_{m=3}^{M} \frac{1 - b_m}{m!},$$

so the carry defects $1 - b_m$ are integer coefficients of a factorial series. Hančl and Tijdeman classify the rational sums with polynomial coefficients [11, Thm. 3.1 and Cor. 3.1, pp. 390–391]; their denominator is the cumulative linear product $\prod_{n \leq N}(an+b)$, and the individual number $N! - 1$ does not occur. In the factorial case $a = 1$, $b = 0$, the criterion of Oppenheim that they reproduce [11, Lem. 2.2, p. 385] says the following: for integer coefficients c_n with $|c_n| < n$ eventually and $\liminf_n |c_n|/n = 0$, the sum $\sum_n c_n/n!$ is rational exactly when $c_n = 0$ eventually. This formulation concerns integer coefficients and factorial denominators, not reciprocal terms with denominator $n! - 1$. However, their introduction records a version of Oppenheim's result without the liminf assumption: under $|c_m| < m - 1$ eventually, rationality is equivalent to eventual vanishing [11, Introduction, p. 383]. Changing finitely many terms adds a rational number, so eventual hypotheses suffice. Here $2 - m \leq c_m = 1 - b_m \leq 2$ gives $|c_m| < m - 1$ for $m \geq 4$; the classical result therefore applies directly.

For completeness, the following elementary proof specialises that result to these one-sided bounds. For $N \geq 2$, telescoping $\sum_{m>N}(m-1)/m! = 1/N!$ gives

$$-1 < N! \sum_{m>N} \frac{c_m}{m!} \leq 2N! \sum_{m>N} \frac{1}{m!} < \frac{2}{N}.$$

The lower bound follows from $c_m \geq 2 - m$ and $N! \sum_{m>N}(m-2)/m! = 1 - N! \sum_{m>N} 1/m! < 1$. The upper bound follows by comparison with $\sum_{j \geq 1}(N+1)^{-j} = 1/N$. If $\sum_{m \geq 3} c_m/m!$ is rational, the scaled tails are integers for all sufficiently large N. They must then be zero, and subtracting consecutive tails gives $c_m = 0$ eventually. The converse is immediate. Since $Z_M/M! \to S$, the displayed finite identity recovers Theorem 4.1. This is a self-contained specialisation of the classical rationality criterion, not a strengthening of it. It proves an equivalence, not the nonvanishing needed for irrationality.

B.3 A superseded deduction

Factorial-congruence multiplicities give another lower bound for L_N. Its exponent is weaker than that of the terminal-block argument in §3. We include the deduction to specify what follows from this external input. For an odd prime r, let m_r count the indices $2 \leq n \leq N$ with $r \mid n! - 1$; such an index satisfies $n < r$. The factorial-congruence

multiplicity estimate of Garaev, Luca and Shparlinski [7, arXiv v1, Thm. 12, p. 16], applied to the residue $a = 1$ on $1 \le n \le \min(N, r-1)$, gives $m_r \ll N^{2/3}$; the prime 2 divides none of these factors. The sum of the exponents at a prime is at most the number of occurrences times the largest exponent, which is its exponent in L_N. Thus

$$\sum_{n=2}^{N} \log(n!-1) = \sum_{r} \sum_{n=2}^{N} v_r(n!-1)\log r \le (\max_r m_r)\log L_N \ll N^{2/3}\log L_N,$$

and Stirling's formula makes the left side $\asymp N^2 \log N$, whence $\log L_N \gg N^{4/3}\log N$. Theorem 3.4 supersedes this: its exponent is $3/2$, its constant is explicit, and it does not use the external multiplicity theorem. The deduction is retained because it is the only place a multiplicity bound enters the record.

Dividing all coefficients by their gcd does not evade the lcm restriction: the resulting integer vector still has $V_2 = \cdots = V_D = 0$, so its value of M is still divisible by L_D. For the nonzero cofactor vector, §1.5 justifies normalisation by identifying the primitive progression vector. It supplies no proof that the remainder is nonintegral. No formalised primitive cofactor construction is claimed here.

B.4 Rational grid points and first crossings

Fix $q > 0$. To compare a partial sum H with a rational number of denominator q, consider the grid $q^{-1}\mathbb{Z}$. Its next point above H is $(\lfloor qH \rfloor + 1)/q$. It is at most S exactly when $\lfloor qH \rfloor + 1 \le qS$. Suppose $H < G \le S$, the number $n!G$ is an integer, and $n!(S-H) < 1$. Then

$$n!H < n!G \le n!S < n!H + 1.$$

Thus $n!G$ is both the least integer strictly above $n!H$ and the floor of $n!S$. In particular, for $n \ge 2$, suppose a single rational level G satisfies $H_{n+1} < G \le S$ and $n!G \in \mathbb{Z}$. Since $H_n < H_{n+1}$, the tail bounds at both indices give

$$\lfloor (n+1)!S \rfloor = (n+1)!G = (n+1)\lfloor n!S \rfloor.$$

The next canonical digit is therefore zero. The same level must persist at both indices; choosing an unrelated rational level at each index would not imply this equality.

Set $H_1 = 0$, the empty partial sum. Now let $\tau \ge 2$ be a first crossing of a rational level G, so that $H_{\tau-1} < G \le H_\tau$, and write $G - H_{\tau-1} = a/v$ with positive integers a, v. The newly added summand gives

$$0 < \frac{a}{v} \le \frac{1}{\tau! - 1}, \qquad v \ge a(\tau! - 1) \ge \tau! - 1.$$

The scaled overshoot is

$$0 \le \tau!(H_\tau - G) = 1 + \frac{1}{\tau! - 1} - \tau!\frac{a}{v} < 2.$$

Consequently $-\lfloor \tau!(H_\tau - G) \rfloor$ is 0 or -1. Here the minus sign is outside the floor; taking the floor of the negative overshoot would give a different integer at nonintegral

overshoots. The displayed integer is -1 precisely when the overshoot is at least 1, or equivalently when

$$\frac{a}{v} \le \frac{1}{\tau!(\tau!-1)}.$$

In that case $v \ge \tau!(\tau!-1)$. This extra small-gap condition is not asserted for every crossing. Nor is $-\lfloor \tau!(H_\tau - G)\rfloor$ automatically the carry b_τ defined from Z_τ: here G is an arbitrary rational level. Neither denominator bound requires a/v to be in lowest terms.

There is also a direct obstruction at prime indices. With Δ_m as defined in §4, the formal source proves for $m \ge 3$ the criterion

$$m \mid Z_m \quad \Longleftrightarrow \quad 1 + \frac{1}{m!-1} < m\Delta_m \le 2 + \frac{1}{m!-1},$$

and if $S = a/q$ with $q > 0$, then $p \mid Z_p$ for every prime $p > q$, so one exact missed prime p gives $q \ge p$. Exact kernel reduction gives $11 \nmid Z_{11}$, hence $q \ge 11$; exact rational normalisation gives $60 \nmid Z_{60}$, $64 \nmid Z_{64}$ and $67 \nmid Z_{67}$, and the prime index 67 gives the checked bound

$$S = \frac{a}{q},\ q > 0 \quad \Longrightarrow \quad q \ge 67.$$

The exact-interval census of the short note replaces 67 by 300000.

The finite geometric-series identity gives another exact decomposition. For a real $x \ne 0, 1$ and an integer $K \ge 0$,

$$\frac{1}{x-1} = \sum_{j=1}^{K} \frac{1}{x^j} + \frac{1}{x^K(x-1)}.$$

When $x = k!$ and a chosen factorial scale is divisible by $(k!)^K$, the scaled finite sum is integral and only the last term retains the factor $k! - 1$ in its denominator. The identity isolates one remaining fraction before exact bounding, and it supplies no cofinal family of nonzero residuals.

The earlier record rejects a proposed divisibility strengthening of the first-crossing bound and reports examples at $m = 52$ and $m = 591$. Its notation for the rational quantity in that claim was not defined, so those reports are not used as a verified counterexample here. The established conclusion remains the lower bound on the denominator's size, not a specified factor dividing it.

There is a second, more arithmetic mechanism at doubled prime indices, stated in (31). The formal theorem is not restricted to individually computed indices; the divisibility criterion holds for the actual partial sums at every odd prime.

B.5 Exact identities for shared prime powers

Let $I \subseteq \{2, 3, \ldots\}$ be finite. We first compute the prime-power levels common to two summand denominators. We then compare those exponents with the ones already supplied by a factorial factor.

Write $D(I) = \operatorname{lcm}_{i<j,\ i,j\in I} \gcd(d_i, d_j)$, with $D(I) = 1$ for an empty or singleton family. For each prime r, its valuation is the second-largest valuation among the d_i, counting missing values as zero; the denominator lcm has the largest valuation. This proves $D(I) \operatorname{lcm}_{i\in I} d_i \mid \prod_{i\in I} d_i$ prime by prime.

The integer $D(I)$ can be updated exactly when one index is added. For the positive factorial-gap denominators, adjoining a new index $a \geq 2$, $a \notin I$, gives

$$D(I \cup \{a\}) = \operatorname{lcm}\Big(D(I),\ \gcd(d_a, \operatorname{lcm}_{j\in I} d_j)\Big),$$

since finite-family gcd and lcm distributivity collapses the lcm of all pairwise gcds against d_a to a single gcd. For a fixed positive integer F, take the lcm with F on both sides:

$$\operatorname{lcm}(F, D(I \cup \{a\})) = \operatorname{lcm}\Big(F, D(I), \gcd(d_a, \operatorname{lcm}_{j\in I} d_j)\Big).$$

Thus one need only retain the denominator lcm and $\operatorname{lcm}(F, D(I))$ at each step. The prescribed factor F is not an extra summand denominator.

There is also an exact bound by the product of the denominators divided by their least common multiple. For a positive integer F, define $\widetilde{C}(I) = \operatorname{lcm}(F, D(I))/F = D(I)/\gcd(F, D(I))$. Since $\widetilde{C}(I)$ divides $D(I)$, the preceding divisibility gives

$$\widetilde{C}(I) \operatorname{lcm}_{j\in I} d_j \mid \prod_{j\in I} d_j, \qquad \widetilde{C}(I) \leq \frac{\prod_{j\in I} d_j}{\operatorname{lcm}_{j\in I} d_j}.$$

For the factorial block this specialises to

$$\widetilde{C}_p \leq \frac{\prod_{n\in I_p}(n!-1)}{\operatorname{lcm}_{n\in I_p}(n!-1)},$$

an upper bound for the shared factor in terms of the denominator product and lcm. The bound alone does not establish (29).

Dividing by F_p subtracts its prime exponents; it does not remove every prime that occurs in F_p. With $F_p = (p-1)!$, $D_p = D(I_p)$ and $C_p = \operatorname{lcm}(F_p, D_p)$, as in the main record,

$$\widetilde{C}_p = \frac{C_p}{F_p} = \frac{D_p}{\gcd(F_p, D_p)}, \qquad v_r(\widetilde{C}_p) = \max\{0, v_r(D_p) - v_r(F_p)\}.$$

For $e > 0$, $r^e \mid \widetilde{C}_p$ exactly when D_p is divisible by $r^{e+v_r(F_p)}$, which in the block forces two distinct denominators to be divisible by that higher power, by the second-largest-valuation formula. If $i < j$ are two such indices and $f = v_r(F_p)$, then $r^{e+f} \mid j!/i! - 1$. Since $r \mid j! - 1$ forces $j < r$, $0 < j!/i! - 1 < r^{j-i}$, hence $e + f < j - i$. For a prime $r \mid \widetilde{C}_p$, this bounds the surviving valuation by $v_r(\widetilde{C}_p) + v_r(F_p) < r$. Since $v_r(\widetilde{C}_p) \geq 1$ and $v_r(F_p) \geq \lfloor (p-1)/r \rfloor$, we obtain $\lfloor (p-1)/r \rfloor \leq r - 2$, hence $p - 1 < r(r-1) < r^2$. Consequently $\widetilde{C}_p$ is coprime to $k!$ whenever $k(k-1) \leq p - 1$: a prime $r \leq k$ dividing both would give $p - 1 < r(r-1) \leq k(k-1) \leq p - 1$. This excludes the small primes but does not bound the product of the remaining prime powers.

If a prime r divides a denominator $n! - 1$ with $n \geq p$, then $r \nmid F_p$ and $r^e \mid \widetilde{C}_p \iff r^e \mid C_p$ for every $e > 0$. This has an exact incidence-count form,

$$r^e \mid \widetilde{C}_p \quad \Longleftrightarrow \quad 1 < \#\{i \in I_p : r^e \mid i! - 1\},$$

so a bound of at most one such index implies $v_r(\widetilde{C}_p) < e$. Counting the exponents gives

$$v_r(\widetilde{C}_p) = \#\{e \in [1, r-1] : 1 < \#\{i \in I_p : r^e \mid i! - 1\}\},$$

and for a prime $r > 2p-1$ the range can be restricted to $e \in [1, 2p-4]$. The same equivalence holds under the exact condition $r \nmid F_p$ in place of $r > 2p-1$, hence in particular for every prime $r \geq p$. At $e = 2$, an upper bound of one on the number of indices gives $v_r(\widetilde{C}_p) \leq 1$.

A surviving prime power also forces two indices to be far apart. If r is prime, $e > 0$ and $r^e \mid \widetilde{C}_p$, there are $i < j$ in I_p with $r^{e+v_r(F_p)} \mid i! - 1$, $r^{e+v_r(F_p)} \mid j! - 1$ and $r^{e+v_r(F_p)} \leq j^{j-i}$; consequently $(2p-1)^d < r^{e+v_r(F_p)}$ forces $d < j-i$. The required spacing follows directly: any two indices $i < j$ with $r^e \mid i! - 1$ and $r^e \mid j! - 1$ satisfy $e < j - i$, because $r \mid j! - 1$ already forces $j < r$ and the inequality for $j!/i! - 1$ then forces the strict separation. More generally, for any two indices $2 \leq i < j$ and any prime power r^e with $e > 0$,

$$r^e \mid i! - 1, \quad r^e \mid j! - 1 \quad \Longrightarrow \quad r^e \leq j^{j-i},$$

so successive solutions of $i! \equiv 1 \pmod{r^e}$ are at least $e + 1$ apart, and

$$(e+1)\,\#\{i \in I_p : r^e \mid i! - 1\} \;\leq\; 2p + e - 2.$$

The resulting bound on the exponent is

$$r \mid \widetilde{C}_p \quad \Longrightarrow \quad v_r(\widetilde{C}_p) + v_r(F_p) < 2p - 3$$

for every prime r and $p \geq 2$. Thus the exponent already present in F_p uses part of the same bound $2p - 3$. To apply (25), one still needs sufficiently strong bounds for the product of the shared prime powers and for the gap ρ_p / R_p.

The available squarefreeness evidence is finite. An exhaustive modular scan through $r \leq 2{,}000{,}000$ and $n \leq 240$ found four pairs (r, n) with $r^2 \mid n! - 1$ and no prime occurring in two such pairs. All 498,501 pairs $2 \leq a < b \leq 1000$ have squarefree $\gcd(a! - 1, b! - 1)$. An aggregate scan through $p = 499$ also reports a ratio below 0.374, but does not specify its logarithmic normalisation; that ratio is not used as a quantitative premise here. These are reported finite computations, not an asymptotic bound or a proof of squarefreeness at all indices.

A finite version of the argument for first prime occurrences compares the product of a chosen set of primes, each at least 5, with $\prod_{2 \leq k \leq B}(k! - 1)$; if the prime product is larger, at least one chosen prime q divides no $k! - 1$ with $2 \leq k \leq B$, while Wilson's theorem bounds its first occurrence by $q - 2$. Wilson reflection limits what a linear-size divisor can be assumed to be: if n is odd, $n < q$, and $q \mid n! - 1$, then $q \mid (q - n - 1)! - 1$. Suppose also that $p \leq n$, both indices lie in I_p, and the reflected index is earlier, equivalently $q < 2n + 1$. The prime q then divides two denominators in the block. Since $q > n \geq p$, it does not divide $F_p = (p-1)!$, so $q \mid \widetilde{C}_p$: removing the factorial factor does not remove this shared prime.

Stewart states that for every $\varepsilon > 0$ there are infinitely many odd n whose least prime factor q of $n! - 1$ satisfies

$$n < q < \left(\frac{\sqrt{145} - 1}{8} + \varepsilon \right) n,$$

estimate (9) of Theorem 1 being stated for $n! + 1$ [8, p. 463] and transferred to $n! - 1$ in the text [8, p. 464]. This controls q relative to the specified index n, not to its first

occurrence. It does not by itself force a repeated divisor. For the minus sign, the bound is already met at all sufficiently large Wilson indices $n = q - 2$. For every prime $q \geq 5$, Wilson gives $q \mid (q-2)! - 1$, and q is its least prime factor. Indeed, every prime divisor exceeds $q - 2$, and $q - 1$ is even; moreover $q/(q-2) \to 1$. The reflected index is then 1, outside the denominators of S.

For example, modulo 11 the only solution of $k! \equiv 1$ with $2 \leq k \leq 9$ is $k = 9$, although $11/9 < (\sqrt{145} - 1)/8$. A small ratio q/n therefore supplies neither a second admissible index nor membership of a reflected index in the chosen block. The earlier reflection argument needs both block-membership hypotheses and distinct indices. Neither the required estimate at first occurrences nor a bound for the total shared part follows from the transferred least-prime-factor estimate alone.

For a selected prime $q \mid R_p$, the factors 1 and q are coprime, and the associated moduli R_p and R_p/q have lcm R_p. Thus this specialisation requires no second selected prime.

The elementary fact behind the projection argument is as follows. Let Z, T, B be nonnegative integers, let $R > 0$, and suppose $Z \equiv T \pmod{R}$. Every positive divisor Q of R with $Z \leq B < Q$ satisfies $T \bmod Q = Z$, since Z is already the least nonnegative representative. Consequently, unequal residues modulo two divisors exceeding B rule out such a Z. More generally, if $T \bmod Q_1 \neq T \bmod Q_2$, then $\min(Q_1, Q_2) \leq T$: otherwise both residues would equal T.

B.6 Finite vectors and exact numerical examples

The three depth-four vectors below all have $M = 1380$. By Theorem 1.3, their remainders differ only by integers. A single finite calculation therefore tests all three:

$$255 + \frac{4}{5} < 242 + 1380 \sum_{d=5}^{8} \frac{1}{d! - 1} < 255 + \frac{5}{6}, \qquad \frac{2 \cdot 1380}{9! - 1} < \frac{1}{100}.$$

The first remainder lies strictly between 255 and 256. Subtracting the corresponding integer offsets gives the other two intervals.

Vector	$\sum_{n>4} z_n$	k	Interval for $\mathcal{R}$
$12K_4 + 253U_6 - 11U_8$	242	-1474	$(255, 256)$
The five-term vector (5)	-44	-1760	$(-31, -30)$
The norm minimizer on $\{2, \dots, 8\}$	-56	-1772	$(-43, -42)$

The second column is the offset from $1380(S - H_4)$; the third is the integer k in $\mathcal{R} = 1380S + k$. Since $1380H_4 = 1716$, subtracting 1716 converts the second column into the third. The five-term remainder is 286 less than the first, and the norm-minimising remainder is another 12 less. All three exclude exactly the same divisors of 1380 by this argument. The coefficient and support improvements in §1.2 do not enlarge that set of exclusions; the larger finite computations in §6 already imply it. The exact offsets are reproduced by `scripts/check_erdos68_channel_moment.py`.

The finite-support vector $\lambda = 2e_3 - e_4$ has, by kernel check, $V_2(\lambda) = 0$, $M = -12$, $V_3(\lambda) = -2$, $V_4(\lambda) = 11$, and $V_d(\lambda) = -12$ for every $d \geq 5$. Under the exact tail

enclosure $1/119 < \sum_{d\geq 5} 1/(d!-1) < 1/50$ its remainder lies strictly between $-93/575$ and $-309/13685$, so it is nonzero and has absolute value less than 1.

In the enlarged coefficient space of §1, exact integer computation verifies the vectors K_D for every $2 \leq D \leq 12$: weighted sums 2 through D vanish, $M = L_D$, and the coefficients have gcd one. They lie outside the space of vectors supported on $n \geq 2$, since L_D is odd whereas M is even for every such vector. At $D = 9$,

$$L_9 = 31540008254514077395, \qquad a_9 = [e_1]K_9 = -3902884074990939115.$$

Since $U_{11} = T_{11}$ has $u_{11} = 0$, the vector $K_9 - 9553024718754\,U_{11}$ keeps the coordinate a_9 and has coefficients with gcd one. By Theorem 1.3 its residual is $L_9(S - H_9) - 9553024718754$, and exact rational arithmetic with the tail bound $\sum_{n\geq 36} 1/(n!-1) < 2/(36!-1)$ places it strictly between $1353/100000$ and $1354/100000$. A different example is supported on $n \geq 2$: the vector $c = (-40, 55, -10, 1)$ on the support $(3, 4, 5, 6)$ annihilates weighted sums 2 and 3, has $M = 600$, and satisfies

$$0.09925341997208298 < \mathcal{R}(c) < 0.09925341997208300,$$

which excludes denominators dividing 600.

A separate interval computation reports the stronger geometric statement that no lower-interval event $m\theta_{m-1} < E_m$ occurs at any $3 \leq m \leq 100000$; its executable and source digest are not available, so that classification remains external finite evidence. The carry computation through 300000 used in the short note is the exact GMP computation recorded in Section 6, with the separate calculation through 4000; it is computational evidence outside the Lean development.

B.7 Limits of the recurrences and clearing factors used here

Two distinctions are needed in using these identities: a recurrence may have solutions that violate the defining floor relations, and a fixed factor may eventually be absorbed by a growing factorial. The examples below keep these cases separate.

- Without the defining floor relation, the carry recurrence and its range allow $Z_m/m!$ to be the constant $3/2$. Taking $Z_m = 3m!/2$ for $m \geq 2$ and $b_m = 1$ for $m \geq 3$ gives $Z_m = mZ_{m-1} + 1 - b_m$ and $-1 \leq b_m \leq m - 1$, with the actual initial value $Z_2 = 3$. But it gives $Z_3 = 9$, whereas the actual partial sum $H_3 = 6/5$ gives $Z_3 = 8$. This example only disproves sufficiency of the stated recurrence, bounds and initial integer. It fails the floor definition. No claim is made that it satisfies the additional prime-index identities. The rational recurrence for $F_m = m!H_m$ in §B.1 determines the actual partial sums uniquely.

- The arguments considered with Wilson quotients, harmonic sums, p-adic gamma identities, and factorial residues still need a real gap estimate and the required modular divisibility at the same indices. No such unbounded family is obtained here. This records the missing step in these arguments, not an impossibility theorem for the use of those identities.

- For the genus-zero product $E(z) = \prod_{n\geq 2}(1 - z/n!)$, local uniform convergence and logarithmic differentiation give $-E'(1)/E(1) = S$. Termwise clearing by $\prod_{n=2}^{N}(n!-$

1) cannot give a positive remainder tending to zero: this product is at least L_N, and $L_N(S - H_N) \to \infty$ by §3. The conclusion concerns this clearing factor alone; Hermite–Padé systems with other denominators are not ruled out.

- Changing a coefficient vector without changing its value of M changes the remainder by an integer only. Also, cancelling the first $D - 1$ weighted sums forces L_D to divide the value of M; factorial divisibility does not remove that constraint.
- A fixed pair of denominator indices cannot make the projection argument work at arbitrarily large parameters, because their factors eventually divide the factorial being removed.

B.8 An elementary criterion for any real number

The fractional-part condition in (20) is an instance of Cantor's termination criterion. For any real x and positive sequence c_m satisfying $c_m \le 1/m$ eventually, we have

$$x \notin \mathbb{Q} \quad \Longleftrightarrow \quad \{(m-1)!\,x\} \ge c_m \text{ for arbitrarily large } m.$$

For rational x the fractional parts vanish eventually. Conversely, if $\{(m-1)!x\} < c_m \le 1/m$ for all large m, then $m\{(m-1)!x\} < 1$, so every sufficiently late canonical digit is zero. Cantor's criterion then makes x rational. Equality $c_m = 1/m$ is allowed: it is failure of the displayed weak inequality that makes $m\{(m-1)!x\}$ strictly less than one. For $x = S$, the choice $c_m = E_m/m$ gives exactly (20); indeed $0 < E_m < 1$ ensures the required threshold bound. The finite test (21) is a separate sufficient condition at one index. Its cofinal converse uses the carry theorem, as shown in §4, not substitution into this general criterion. Neither argument proves that S meets the test at arbitrarily large indices.

The threshold $1/m$ cannot be replaced by λ/m for a fixed $\lambda > 1$. For $x = e$ and $m \ge 3$, the factorial tail gives

$$m\{(m-1)!e\} = 1 + m! \sum_{n>m} \frac{1}{n!} < 1 + \frac{1}{m} < \lambda \qquad \text{eventually.}$$

Thus the irrational number e would never meet the enlarged threshold at sufficiently large indices. Positivity is also essential: with $c_m = 0$, rational numbers would meet the threshold eventually. These examples explain the two assumptions without imposing any equidistribution hypothesis.

C Relations among the criteria

The classification determines all cancellation vectors and attainable multipliers. The progression construction adds eventual divisibility by each fixed q and a vanishing tail bound for the same vector. Its primitive cofactor construction gives that vector up to sign, not another family. At fixed M, coefficient changes still alter only the integer in $\mathcal{R}(c) = M(c)S + k$.

Common-denominator growth measures the cost of clearing, while prime-power survival describes reduction. Neither supplies the missing gap comparison. The carry,

digit and interval criteria characterize irrationality but leave their required unbounded departures unproved. The finite exclusions restrict divisibility and size, respectively; they establish no unbounded assertion.

The remaining target for this method is therefore precise: for every $q > 0$, find a nonintegral remainder whose own multiplier satisfies $q \mid M$. The algebraic classification supplies choices, not this arithmetic conclusion.

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An AI research pass supplied by Will Cook derived the depth-four dual congruence and shorter vectors from the weighted linear forms and the classification of their possible coefficients M developed earlier here. OpenAI Codex checked and integrated them; a separate AI pass reviewed the proof.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.1	Lean · Comparator	Theorem 1.3	Lean · Comparator
Theorem 1.4	Lean · Comparator	Theorem 1.2	Lean · Comparator
Theorem 1.7	Lean · Comparator	Theorem 2.1	Lean · Comparator
Proposition 2.2	Lean · Comparator	Lemma 3.1	Lean · Comparator
Lemma 3.2	Lean · Comparator	Lemma 3.3	Lean · Comparator
Theorem 3.4	Lean · Comparator	Theorem 1.5	Lean · Comparator
Corollary 1.6	Lean · Comparator	Theorem 4.1	Lean · Comparator
Proposition 4.2	Lean · Comparator	Proposition 4.3	Lean · Comparator
Theorem 4.4	Lean · Comparator	Theorem 5.1	Lean · Comparator